Revision of the Grade 7 course

Angles on parallel lines, the angle sum of a triangle, isosceles triangles and the congruence tests — the facts every proof this quarter will quote.

LESSON 1–2 | 2 × 40 MIN | GEOMETRY 8, P. 3 | STAGE 9 · 5.1

LESSON CLOCK

5' | 14' | 7' | 12' | 2'

Warm-up	5 min
Explanation	14 min
Interactive	7 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Use vertically opposite, corresponding, alternate and co-interior angles.
- Use the angle sum of a triangle and the exterior angle theorem.
- State and use the properties of an isosceles triangle.
- Quote the four congruence tests correctly (SSS, SAS, ASA, RHS).

TEXTBOOK

National	pp. 3–4
Cambridge	Learner’s Book pp. 104–108
Workbook	Workbook 5.1

01 Explanation

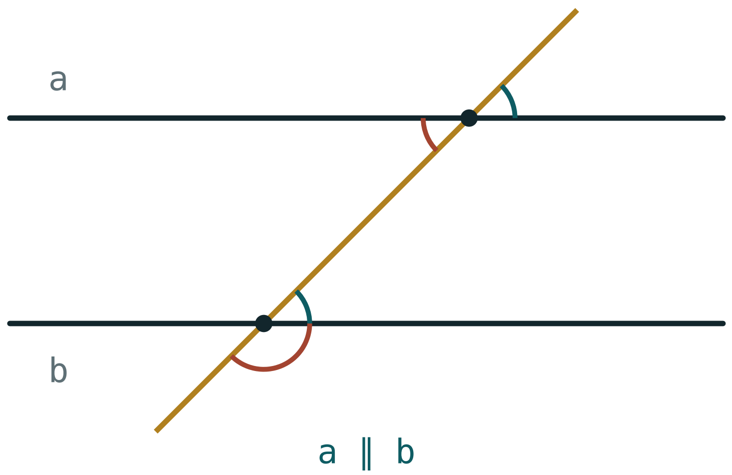
Why these four facts

Every proof in Chapter I — that the opposite sides of a parallelogram are equal, that the diagonals of a rectangle are equal, that the midline is half the base — is built from the same small set of Grade 7 facts. Two lessons spent making them automatic pays for the whole quarter.

1 · Angles on parallel lines

When a transversal crosses two parallel lines:

- **Corresponding** angles are equal (the “F” shape).
- **Alternate** angles are equal (the “Z” shape).
- **Co-interior** angles add to $180°$ (the “C” shape).



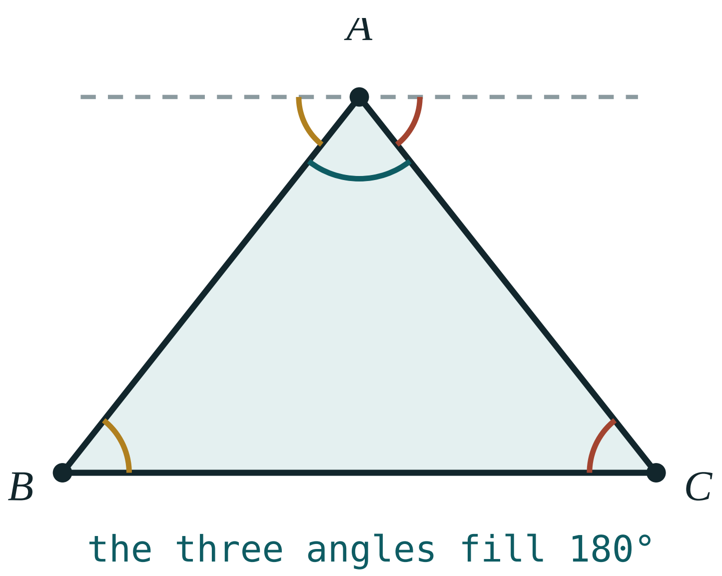
A transversal crossing two parallel lines. Equal angles are marked with the same colour.

And independently of any parallel lines: **vertically opposite** angles are equal, and angles on a straight line add to $180°$.

2 · The angle sum of a triangle

THEOREM

The three angles of any triangle add up to $180°$.



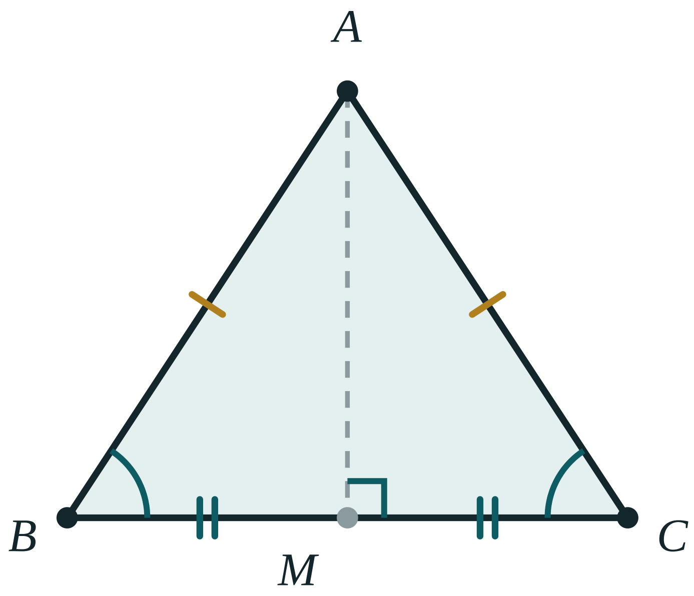
Draw a line through A parallel to BC. The two outer angles equal ∠B and ∠C (alternate angles), and together with ∠A they fill a straight line.

That figure *is* the proof: the dashed line through A is parallel to BC, so the two outer angles equal $\angle B$ and $\angle C$, and the three angles together make a straight line.

COROLLARY — THE EXTERIOR ANGLE

An exterior angle of a triangle equals the sum of the two interior angles not next to it.

3 · The isosceles triangle

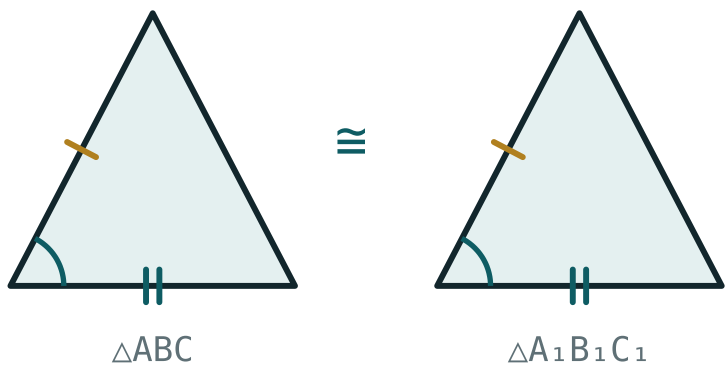


$$AB = AC \implies \angle B = \angle C$$

In an isosceles triangle the base angles are equal, and the line from the apex to the midpoint of the base is at the same time the median, the height and the angle bisector.

- $AB = AC \implies \angle B = \angle C$ — and the converse is also true.
- The median from the apex is also the height and the bisector of the apex angle.
- An equilateral triangle has all three angles $60°$.

4 · The congruence tests



Two triangles with two equal sides and the equal angle between them — the SAS test.

1. **SSS** — three sides.
2. **SAS** — two sides and the angle *between* them.
3. **ASA** — two angles and the side between them.
4. **RHS** — right angle, hypotenuse and one other side.

THERE IS NO “SSA”

Two sides and an angle that is *not* between them does not fix the triangle. Learners quote this constantly; correct it now rather than in the control work.

In congruent triangles, corresponding sides and corresponding angles are equal. Naming the vertices in the right order — $\triangle ABC \cong \triangle A_1B_1C_1$ — is what tells the reader which parts correspond.

02 Worked examples

EXAMPLE 1 ***In $\triangle ABC$, $\angle A = 52°$ and $\angle B = 61°$. Find $\angle C$ and the exterior angle at C.***

1. $\angle A + \angle B + \angle C = 180°$

   Angle sum of a triangle.

2. $\angle C = 180° - 52° - 61° = 67°$

   Subtract.

3. $exterior\ angle = 180° - 67° = 113°$

   Angles on a straight line.

4. $check{:}\ 52° + 61° = 113°$ ✓

   The exterior angle equals the two opposite interior angles.

ANSWER

$\angle C = 67°$, exterior angle $113°$

EXAMPLE 2 ***$\triangle ABC$ is isosceles with $AB = AC$ and $\angle A = 40°$. Find $\angle B$.***

1. $\angle B = \angle C$

   Base angles of an isosceles triangle.

2. $40° + 2\angle B = 180°$

   Angle sum, with ∠C replaced by ∠B.

3. $2\angle B = 140°$

4. $\angle B = 70°$

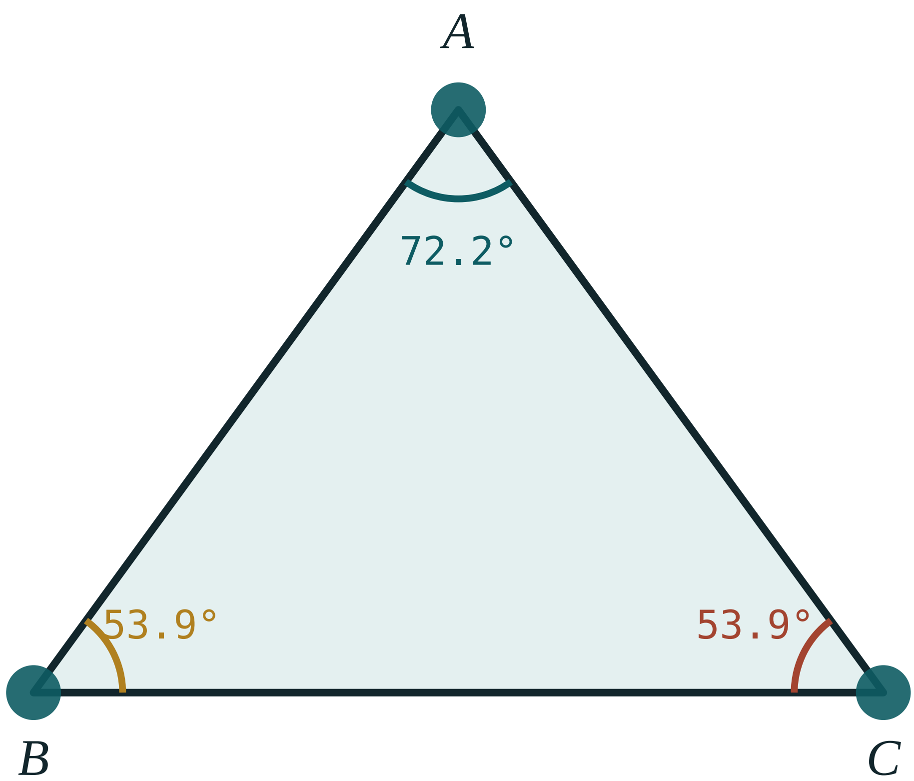
ANSWER

$\angle B = \angle C = 70°$

03 Interactive model

Drag a vertex until the triangle is nearly flat and ask what happens to the sum. It stays 180°.

The angles of a triangle



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Transversal	Kesuvchi	Секущая
Corresponding angles	Mos burchaklar	Соответственные углы
Alternate angles	Almashinuvchi burchaklar	Накрест лежащие углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Isosceles triangle	Teng yonli uchburchak	Равнобедренный треугольник
Congruent	Teng (mos)	Равные
Median	Mediana	Медиана
Bisector	Bissektrisa	Биссектриса
Altitude (height)	Balandlik	Высота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Alternate angles on parallel lines are:

- equal
- supplementary
- complementary
- unrelated

Co-interior angles on parallel lines add to:

- $90°$
- $180°$
- $360°$
- they are equal

An exterior angle of a triangle equals:

- the nearest interior angle
- the sum of the two opposite interior angles
- $180°$
- half the angle sum

Which is NOT a congruence test?

- SSS
- SAS
- SSA
- ASA

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Two angles of a triangle are 40° and 75°. Find the third.*

   ANSWER *65°*

2. *An isosceles triangle has apex angle 40°. Find each base angle.*

   ANSWER *70°*

3. *Two parallel lines are cut by a transversal. One angle is 65°. Find its corresponding angle.*

   ANSWER *65°*

4. *Find the co-interior partner of an angle of 110°.*

   ANSWER *70°*

5. *Find the angle vertically opposite an angle of 128°.*

   ANSWER *128°*

6. *Each angle of an equilateral triangle is:*

   ANSWER *60°*

7. *An exterior angle of a triangle is 120°. Find the adjacent interior angle.*

   ANSWER *60°*

Medium · the standard question

7 PROBLEMS

1. *In $\triangle ABC$, $\angle A = 52°$ and $\angle B = 61°$. Find the exterior angle at C.*

   ANSWER $113°$

2. *An isosceles triangle has a base angle of 35°. Find the apex angle.*

   ANSWER $110°$

3. *The angles of a triangle are in the ratio 2 : 3 : 4. Find them.*

   ANSWER $40°, 60°, 80°$

4. *In $\triangle ABC$, $AB = AC$ and $\angle A = 2\angle B$. Find all three angles.*

   ANSWER $\angle A = 90°, \angle B = \angle C = 45°$

5. *Two angles of a triangle are equal and the third is 96°. Find them.*

   ANSWER $42°$ each

6. *Name the congruence test: two triangles share two sides and the angle between them.*

   ANSWER SAS

7. *Name the congruence test: two right triangles have equal hypotenuse and one equal leg.*

   ANSWER RHS

Hard · stretch and reason

7 PROBLEMS

1. *In $\triangle ABC$ the exterior angle at A is $130°$ and $\angle B = 55°$. Find $\angle C$.*

   ANSWER $\angle C = 75°$ — the exterior angle equals $\angle B + \angle C$.

2. *$\triangle ABC$ is isosceles with $AB = AC$. The bisector of $\angle B$ meets AC at D and $\angle ABD = 25°$. Find $\angle A$.*

   ANSWER $\angle B = 50°$, so $\angle C = 50°$ and $\angle A = 80°$

3. *Prove that the median to the base of an isosceles triangle is also its height.*

   ANSWER The two half-triangles are congruent by SSS, so the two angles at the base of the median are equal and together make $180°$ — each is $90°$.

4. *In $\triangle ABC$, $\angle A = 90°$ and $\angle B = \angle C$. Find $\angle B$, and say what kind of triangle it is.*

   ANSWER $45°$; a right isosceles triangle.

5. *Two parallel lines are cut by a transversal so that one pair of co-interior angles is in the ratio $4 : 5$. Find both.*

   ANSWER $80°$ and $100°$

6. *Explain why two triangles with equal angles need not be congruent.*

   ANSWER Equal angles fix the *shape* only. The triangles are similar; without one equal side they can be any size.

7. *$\triangle ABC$ has $\angle A = \angle B$. Prove that $CA = CB$.*

   ANSWER Converse of the isosceles-triangle theorem: drop the bisector of $\angle C$; the two triangles are congruent by ASA, so $CA = CB$.

07 Homework

Homework — 5 problems

Geometry 8, pp. 3–4 (revision). Draw a figure for every question.

1. *Two angles of a triangle are $38°$ and $47°$. Find the third and the exterior angle at that vertex.*
2. *An isosceles triangle has apex angle $96°$. Find the base angles.*

3. *The angles of a triangle are in the ratio 1 : 2 : 3. Find them and name the triangle.*
4. *A transversal cuts two parallel lines. One co-interior angle is 3 times the other. Find both.*
5. Write out the four congruence tests, each with a sketch.

Home / Grade 8 / Geometry / Quarter 1

Polygons. Interior and exterior angles

Splitting a polygon into triangles gives $180°(n-2)$ — and the exterior angles always add to 360°, however many sides there are.

LESSON 3–4 | 2 × 40 MIN | GEOMETRY 8, TEMA 1 | STAGE 9 · 5.2–5.3

LESSON CLOCK

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define a polygon, its vertices, sides, diagonals and angles.
- Distinguish convex from non-convex polygons.
- Derive and use $180°(n-2)$ for the sum of the interior angles.
- Use the fact that the exterior angles of any convex polygon add to $360°$.

TEXTBOOK

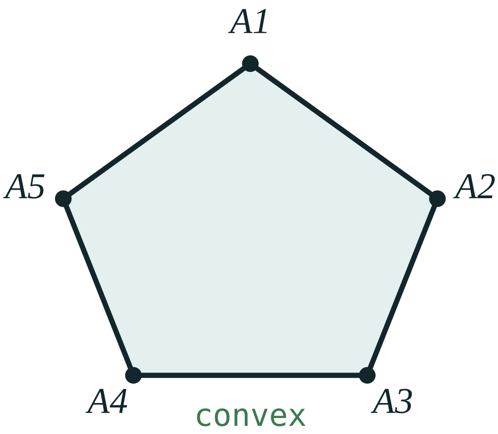
National	Тема 1, pp. 5–7
Cambridge	Learner’s Book pp. 109–118
Workbook	Workbook 5.2–5.3

01 Explanation

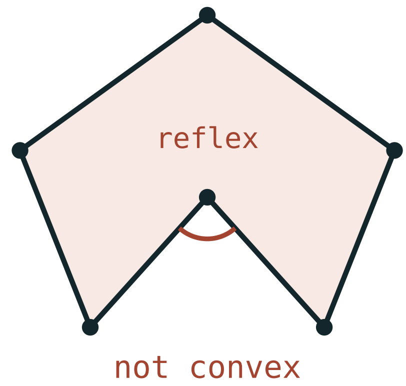
What a polygon is

DEFINITION

A **polygon** is a closed broken line together with the part of the plane it encloses. Its corners are **vertices**, its segments are **sides**, and a segment joining two non-adjacent vertices is a **diagonal**. A polygon with n sides is an n-gon and has n vertices and n angles.



Convex — the whole polygon lies on one side of every side-line.



Not convex — one interior angle is reflex, and the polygon crosses the line of a side.

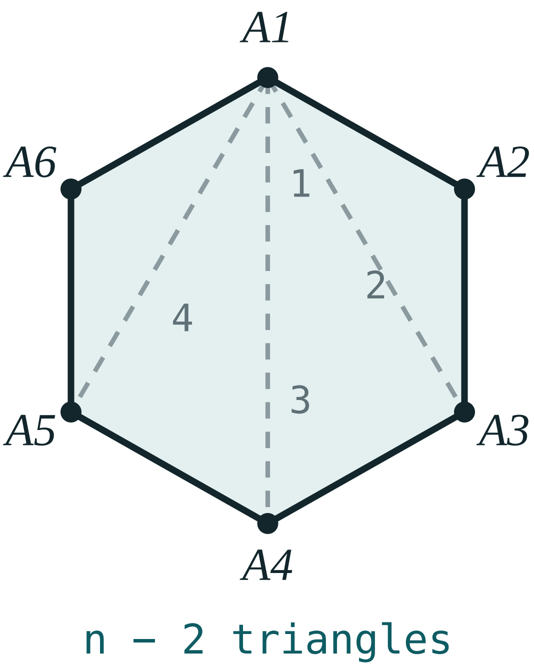
A polygon is **convex** if it lies entirely on one side of the line containing any of its sides. Equivalently: every interior angle is less than $180°$. Every triangle is convex.

The sum of the interior angles

THEOREM

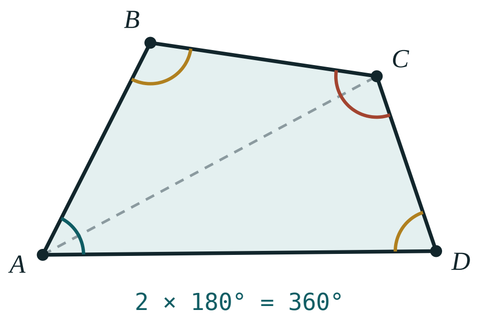
The interior angles of a convex n-gon add up to

$$180° \cdot (n - 2)$$



All the diagonals from one vertex split the hexagon into $4 = 6 - 2$ triangles.

Proof. Choose one vertex and draw all the diagonals from it. They cut the n-gon into exactly $n - 2$ triangles — one fewer than the number of sides you can "see" from that vertex. The angles of all these triangles together make up exactly the angles of the polygon, so the sum is $180°(n - 2)$. ∎



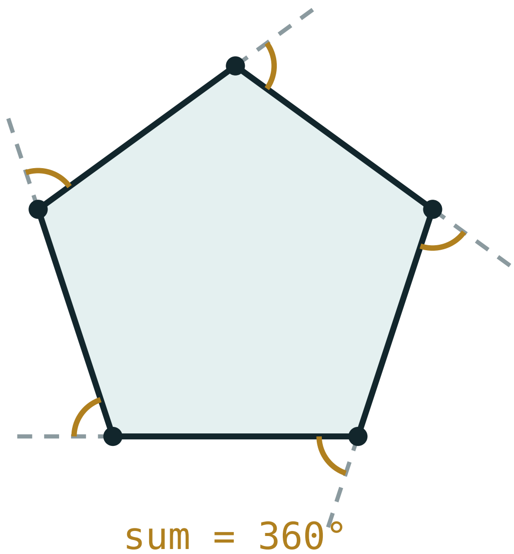
A quadrilateral is two triangles — 2 × 180° = 360°.

POLYGON	N	TRIANGLES	ANGLE SUM	EACH ANGLE IF REGULAR
Triangle	3	1	180°	60°
Quadrilateral	4	2	360°	90°
Pentagon	5	3	540°	108°
Hexagon	6	4	720°	120°
Octagon	8	6	1080°	135°

The exterior angles

THEOREM

Taking one exterior angle at each vertex, the exterior angles of any convex polygon add up to $360°$ — **whatever the number of sides**.



One exterior angle at each vertex. Walking once round the polygon turns you through a full circle.

Why. At each vertex, interior + exterior = $180°$. Over n vertices that is $180°n$ altogether, and the interior angles account for $180°(n - 2) = 180°n - 360°$. What is left for the exterior angles is exactly $360°$.

The walking picture is worth showing: set off along one side and walk right round the polygon. At each corner you turn by the exterior angle, and when you arrive back at the start you are facing your original direction — you have turned through one full revolution.

For a **regular** n-gon this makes the arithmetic immediate: each exterior angle is $\frac{360°}{n}$, so each interior angle is $180° - \frac{360°}{n}$.

02 Worked examples

EXAMPLE 1 **Find the sum of the interior angles of a decagon, and each angle if it is regular.**

1. $n = 10$

   A decagon has ten sides.

2. $180°(10 - 2) = 180° \cdot 8 = 1440°$

   Apply the formula.

3. $\frac{1440°}{10} = 144°$

   Regular means all ten angles are equal.

4. check: exterior angle $= \frac{360°}{10} = 36°$, and $180° - 36° = 144°$ ✓

   Two routes, same answer.

ANSWER

$1440°$; each angle $144°$

EXAMPLE 2 **The interior angles of a convex polygon add up to 1260°. How many sides has it?**

1. $180°(n - 2) = 1260°$

   Set up the equation.

2. $n - 2 = 7$

   Divide by 180.

3. $n = 9$

   A nonagon.

ANSWER

9 sides

EXAMPLE 3 **Each interior angle of a regular polygon is 156°. Find the number of sides.**

1. $\text{exterior angle} = 180° - 156° = 24°$

   Work with the exterior angle — far quicker.

2. $n \cdot 24° = 360°$

   All exterior angles are equal and add to 360°.

3. $n = 15$

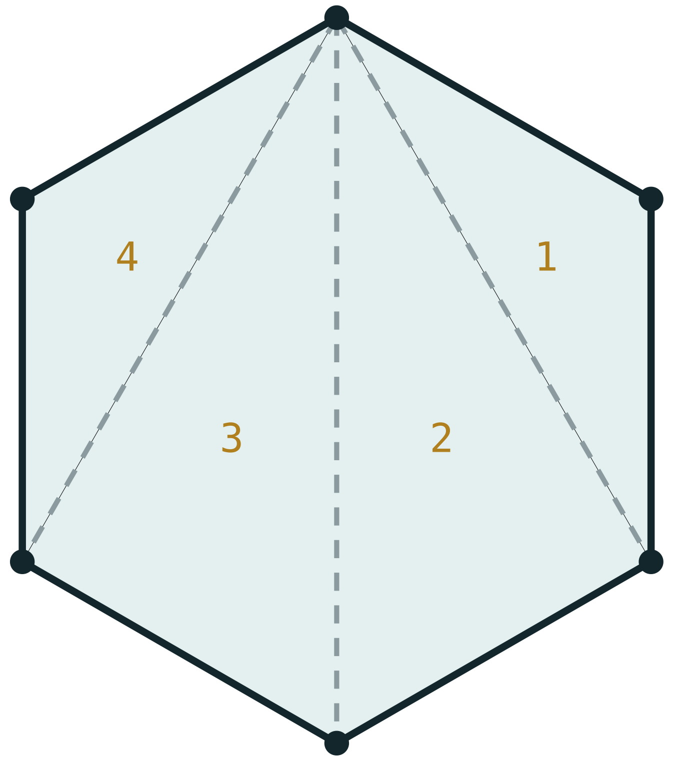
ANSWER

15 sides

03 Interactive model

Slide n from 3 to 12 and let the class read off the triangle count before the formula appears.

Angle sum of a polygon



$$180° \times (6 - 2) = 720°$$

sides n **6** | triangles **4** | interior sum **720°** | each angle **120°** | exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Polygon	Ko‘pburchak	Многоугольник
Convex	Qavariq	Выпуклый
Side	Tomon	Сторона
Diagonal	Diagonal	Диагональ
Interior angle	Ichki burchak	Внутренний угол
Exterior angle	Tashqi burchak	Внешний угол
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Sum of the angles	Burchaklar yig‘indisi	Сумма углов
Pentagon	Beshburchak	Пятиугольник
Hexagon	Oltiburchak	Шестиугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The interior angles of a hexagon add up to:

- $540°$
- $720°$
- $900°$
- $360°$

The exterior angles of a convex 20-gon add up to:

- $360°$
- $3240°$
- $720°$
- $1800°$

Each exterior angle of a regular polygon is 45°. It has:

- 6 sides
- 8 sides
- 9 sides
- 12 sides

A polygon is convex when:

- all its sides are equal
- every interior angle is less than 180°
- it has an even number of sides
- its diagonals are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Sum of the interior angles of a pentagon*

   ANSWER *540°*

2. *Sum of the interior angles of an octagon*

   ANSWER *1080°*

3. *Each interior angle of a regular hexagon*

   ANSWER *120°*

4. *Each exterior angle of a regular pentagon*

   ANSWER *72°*

5. *Sum of the exterior angles of a convex 12-gon*

   ANSWER *360°*

6. *How many triangles do the diagonals from one vertex of a heptagon make?*

   ANSWER *5*

7. *Three angles of a quadrilateral are 80°, 95° and 100°. Find the fourth.*

   ANSWER *85°*

Medium · the standard question

7 PROBLEMS

1. *The interior angles of a convex polygon add to 1260°. Find n.*

   ANSWER $n = 9$

2. *Each interior angle of a regular polygon is 156°. Find n.*

   ANSWER $n = 15$

3. *Each exterior angle of a regular polygon is 24°. Find n and the interior angle.*

   ANSWER $n = 15$, interior $156°$

4. *Sum of the interior angles of a regular decagon, and each angle*

   ANSWER $1440°$, $144°$

5. *The angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4. Find them.*

   ANSWER $36°$, $72°$, $108°$, $144°$

6. *A convex polygon has 20 sides. Find the sum of its interior angles.*

   ANSWER $3240°$

7. *How many diagonals can be drawn from one vertex of an n-gon?*

   ANSWER $n - 3$

Hard · stretch and reason

7 PROBLEMS

1. *Each interior angle of a regular polygon is 4 times its exterior angle. Find n.*

   ANSWER exterior = $36°$, so $n = 10$

2. *The interior angles of a convex polygon add to 2340°. Find n.*

   ANSWER $n = 15$

3. *Show that a convex polygon can have at most three acute interior angles.*

   ANSWER Four or more acute angles would leave four exterior angles each above $90°$, already exceeding $360°$.

4. *Find the total number of diagonals of an n-gon.*

   ANSWER $\frac{n(n-3)}{2}$ — each vertex sends $n - 3$, and each diagonal is counted twice.

5. *Is a regular polygon with each interior angle 155° possible? Explain.*

   ANSWER No — the exterior angle would be $25°$ and $360 \div 25 = 14.4$ is not a whole number.

6. *The angles of a pentagon are 5 consecutive whole numbers of degrees. Find them.*

   ANSWER $106°, 107°, 108°, 109°, 110°$

7. *A regular polygon has interior angle 3 times that of a regular hexagon minus 180°. Find n.*

   ANSWER interior = $3 \cdot 120° - 180° = 180°$ — impossible, so no such polygon exists.

07 Homework

Homework — 5 problems

Geometry 8, Тема 1, pp. 5–7. Cambridge Learner's Book 5.2–5.3 for extra practice.

1. *Find the sum of the interior angles of a 14-gon.*
2. *Each interior angle of a regular polygon is 162°. Find the number of sides.*
3. *Three angles of a quadrilateral are equal and the fourth is 120°. Find them.*
4. *A regular polygon has exterior angle 20°. Find n and the interior angle.*

5. *Draw a convex and a non-convex hexagon, and mark the reflex angle on the second.*

Parallelogram and its properties

One definition — both pairs of opposite sides parallel — and three properties that follow from a single pair of congruent triangles.

LESSON 5 | 1 × 40 MIN | GEOMETRY 8, TEMA 2 | STAGE 9 · BEYOND

LESSON CLOCK

4' | 13' | 8' | 13' | 2'

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of a parallelogram.
- Prove and use: opposite sides equal, opposite angles equal, diagonals bisect each other.
- Use the fact that neighbouring angles are supplementary.
- Solve numerical problems on sides, angles and perimeter.

TEXTBOOK

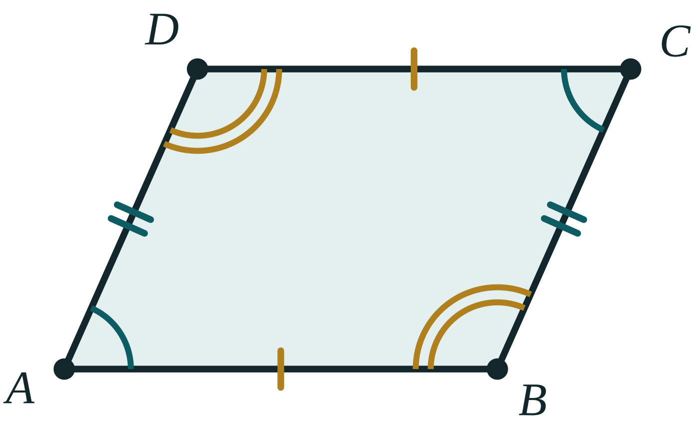
National	Тема 2, pp. 8–10
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The definition

DEFINITION

A **parallelogram** is a quadrilateral whose opposite sides are pairwise parallel: $AB \parallel DC$ and $AD \parallel BC$.



Opposite sides carry the same tick marks; opposite angles carry the same arcs. Neighbouring angles add to 180°.

Notice what the definition does *not* say. It says nothing about lengths and nothing about angles. Everything else has to be proved.

Property 1 — opposite sides are equal

Proof. Draw the diagonal AC. It cuts the parallelogram into $\triangle ABC$ and $\triangle CDA$.

- AC is common to both.
- $\angle BAC = \angle DCA$ — alternate angles, since $AB \parallel DC$.
- $\angle BCA = \angle DAC$ — alternate angles, since $AD \parallel BC$.

So $\triangle ABC \cong \triangle CDA$ by ASA, and therefore $AB = CD$ and $BC = AD$. ∎

CONSEQUENCE

Perimeter $P = 2(a + b)$, where a and b are two neighbouring sides.

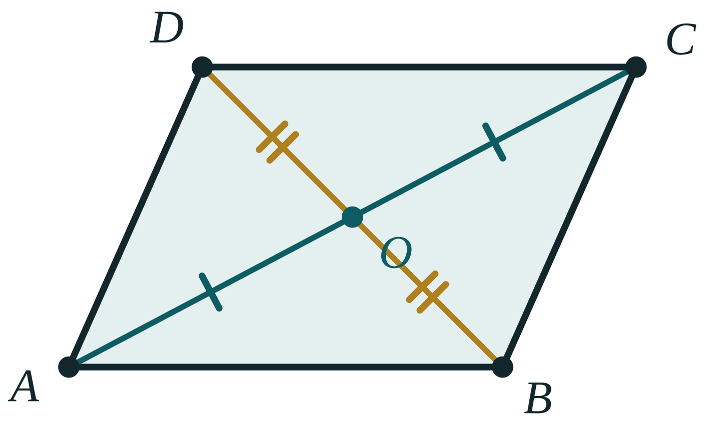
Property 2 — opposite angles are equal

The same congruent triangles give $\angle B = \angle D$ directly. For $\angle A$ and $\angle C$, add the two pairs of alternate angles at each of them.

AND NEIGHBOURING ANGLES

$\angle A + \angle B = 180°$ — they are co-interior angles between the parallels AD and BC. So knowing **one** angle of a parallelogram gives you all four.

Property 3 — the diagonals bisect each other



The diagonals cross at O, and each is cut into two equal parts: AO = OC and BO = OD.

Proof. Let the diagonals meet at O. In $\triangle AOB$ and $\triangle COD$:

- $AB = CD$ — Property 1.
- $\angle OAB = \angle OCD$ and $\angle OBA = \angle ODC$ — alternate angles on $AB \parallel DC$.

So $\triangle AOB \cong \triangle COD$ by ASA, giving $AO = OC$ and $BO = OD$. ∎

BISECT, NOT EQUAL

The diagonals cut **each other** in half. They are **not** equal to one another — that only happens in a rectangle, and it is the single most common error in this chapter.

02 Worked examples

EXAMPLE 1 ***In parallelogram ABCD, $\angle A = 65°$. Find the other three angles.***

1. $\angle C = \angle A = 65°$

   Opposite angles are equal.

2. $\angle B = 180° - 65° = 115°$

   Neighbouring angles are supplementary.

3. $\angle D = \angle B = 115°$

   Opposite angles again.

4. *check:* $65 + 115 + 65 + 115 = 360$ ✓

   The angle sum of a quadrilateral.

ANSWER

$\angle B = \angle D = 115°, \angle C = 65°$

EXAMPLE 2

The perimeter of a parallelogram is 48 cm and one side is 3 cm longer than the other. Find the sides.

1. $2(a + b) = 48, \text{ so } a + b = 24$

   Perimeter formula.

2. $b = a + 3$

   The given relation.

3. $a + a + 3 = 24 \Longrightarrow 2a = 21 \Longrightarrow a = 10.5$

4. $b = 13.5$

ANSWER

$10.5\ cm$ and $13.5\ cm$

EXAMPLE 3

The diagonals of parallelogram ABCD meet at O, with AC = 14 cm and BD = 10 cm. Find AO and OD.

1. $AO = OC = \frac{14}{2} = 7\ cm$

   The diagonals bisect each other.

2. $BO = OD = \frac{10}{2} = 5\ cm$

   Same property, other diagonal.

3. Note that $AC \neq BD$ — the diagonals are not equal here.

   They would be only in a rectangle.

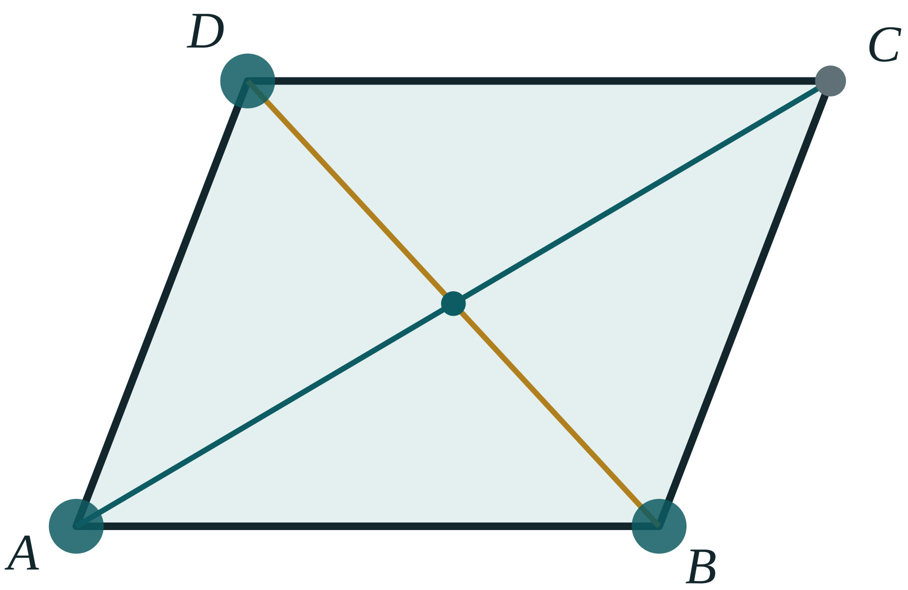
ANSWER

$AO = 7\ cm, OD = 5\ cm$

03 Interactive model

Drag A, B or D and watch AB = DC and AD = BC hold, while AC and BD stay different.

Drag the parallelogram



AB 170 | DC 170 | AD 139.3 | BC 139.3 | ∠A 69° | ∠C 69° | AC 255.5

BD 176.9 | AO 127.8 | OC 127.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Parallelogram	Parallelogramm	Параллелограмм
Opposite sides	Qarama-qarshi tomonlar	Противоположные стороны
Opposite angles	Qarama-qarshi burchaklar	Противоположные углы
Adjacent angles	Qo‘shni burchaklar	Прилежащие углы
The diagonals bisect each other	Diagonallar bir-birini teng ikkiga bo‘ladi	Диагонали делятся пополам
Perimeter	Perimetr	Периметр
Property	Xossa	Свойство
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In a parallelogram, the diagonals:

- are equal
- bisect each other
- are perpendicular
- bisect the angles

$\angle A = 110°$ in parallelogram ABCD. Then $\angle B$ is:

- $110°$
- $70°$
- $55°$
- $90°$

A parallelogram has sides 7 cm and 5 cm. Its perimeter is:

- $12\ cm$
- $24\ cm$
- $35\ cm$
- $17\ cm$

The proof that opposite sides are equal uses:

- SSS
- ASA with a diagonal
- RHS
- the angle sum of a quadrilateral

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *In parallelogram ABCD, AB = 8 cm. Find CD.*

   ANSWER *8 cm*

2. $\angle A = 70°$. *Find* $\angle C$.

   ANSWER *70°*

3. $\angle A = 70°$. *Find* $\angle B$.

   ANSWER *110°*

4. *Sides 6 cm and 9 cm. Find the perimeter.*

   ANSWER *30 cm*

5. *AC = 12 cm and the diagonals meet at O. Find AO.*

   ANSWER *6 cm*

6. *BD = 9 cm. Find OD.*

   ANSWER *4.5 cm*

7. *One angle of a parallelogram is 90°. What are the others?*

   ANSWER *90°* each — it is a rectangle.

Medium · the standard question

7 PROBLEMS

1. $\angle A = 65°$. *Find all four angles.*

   ANSWER $65°, 115°, 65°, 115°$

2. *The perimeter is 48 cm and one side is 3 cm longer than the other. Find the sides.*

   ANSWER *10.5 cm* and *13.5 cm*

3. $AC = 14$ *cm*, $BD = 10$ *cm. Find AO and OD.*

   ANSWER $AO = 7$ *cm*, $OD = 5$ *cm*

4. *Two neighbouring angles are in the ratio 2 : 3. Find all four angles.*

   ANSWER $72°, 108°, 72°, 108°$

5. *The perimeter is 36 cm and one side is twice the other. Find the sides.*

   ANSWER *6 cm* and *12 cm*

6. $\angle A - \angle B = 40°$. *Find both angles.*

   ANSWER $\angle A = 110°, \angle B = 70°$

7. *In parallelogram ABCD,* $AO = 5$ *cm and* $OD = 3$ *cm. Find AC and BD.*

   ANSWER $AC = 10$ *cm*, $BD = 6$ *cm*

Hard · stretch and reason

7 PROBLEMS

1. *The bisector of $\angle A$ of parallelogram ABCD meets BC at K. Prove that $BK = AB$.*

   ANSWER $\angle BAK = \angle KAD = \angle AKB$ (alternate angles), so $\triangle ABK$ is isosceles and $BK = AB$.

2. *In parallelogram ABCD, $AB = 6$, $BC = 10$, and the bisector of $\angle A$ meets BC at K. Find KC.*

   ANSWER $BK = AB = 6$, so $KC = 10 - 6 = 4$.

3. *Prove that a parallelogram with one right angle is a rectangle.*

   ANSWER Neighbouring angles are supplementary, so the neighbours are $90°$ too, and opposite angles equal them.

4. *The diagonals of a parallelogram are 16 cm and 12 cm and one side is 10 cm. Find the perimeter, given the diagonals are perpendicular.*

   ANSWER Perpendicular diagonals make it a rhombus, so all sides are $10\ cm$ and $P = 40\ cm$.

5. *In parallelogram ABCD the perpendicular from B to AD has length 6 and $AD = 9$. Find the area.*

   ANSWER $S = 9 \cdot 6 = 54$ square units.

6. *Prove that if the diagonals of a quadrilateral bisect each other, the opposite sides are equal.*

   ANSWER $\triangle AOB \cong \triangle COD$ by SAS (vertically opposite angles), so $AB = CD$; similarly $BC = AD$.

7. *In parallelogram ABCD, $\angle A$ is three times $\angle B$. Find all four angles.*

   ANSWER $\angle A = \angle C = 135°, \angle B = \angle D = 45°$

07 Homework

Homework — 5 problems

Geometry 8, Тема 2, pp. 8–10. Draw and label a figure for each.

1. *In parallelogram ABCD, $\angle D = 118°$. Find the other three angles.*
2. *The perimeter is 54 cm and one side is 5 cm shorter than the other. Find the sides.*
3. *The diagonals meet at O with $AO = 6.5$ cm and $BO = 4$ cm. Find AC and BD.*

4. *Two neighbouring angles are in the ratio 4 : 5. Find all four.*

5. *Prove that the diagonal of a parallelogram divides it into two congruent triangles.*

Tests for a parallelogram

The properties run backwards: three conditions, any one of which is enough to guarantee that a quadrilateral is a parallelogram.

LESSON 6 | 1 × 40 MIN | GEOMETRY 8, TEMA 3 | STAGE 9 · BEYOND

LESSON CLOCK

4' | 14' | 7' | 13' | 2'

Warm-up	4 min
Explanation	14 min
Interactive	7 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the three tests for a parallelogram.
- Tell a property (given a parallelogram, deduce …) from a test (given …, deduce a parallelogram).
- Prove a quadrilateral is a parallelogram using the appropriate test.
- Recognise a condition that is *not* sufficient.

TEXTBOOK

National	Тема 3, pp. 11–13
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Property or test?

Last lesson every statement began "*in a parallelogram* …". This lesson every statement ends "*… therefore it is a parallelogram*". Same facts, opposite direction — and getting the direction right is most of the marks in a proof.

PROPERTY (TEMA 2)	TEST (TEMA 3)
A parallelogram has opposite sides equal.	Opposite sides equal $\Longrightarrow$ parallelogram.
A parallelogram has opposite angles equal.	Opposite angles equal $\Longrightarrow$ parallelogram.
Its diagonals bisect each other.	Diagonals bisecting each other $\Longrightarrow$ parallelogram.

Test 1 — two opposite sides equal and parallel

TEST 1

If two **opposite** sides of a quadrilateral are equal **and** parallel, it is a parallelogram.

Proof. Suppose $AB = CD$ and $AB \parallel CD$. Draw the diagonal AC. Then $\angle BAC = \angle DCA$ (alternate angles) and AC is common, so $\triangle ABC \cong \triangle CDA$ by SAS. Hence $\angle BCA = \angle DAC$, and those are alternate angles for BC and AD — so $BC \parallel AD$ and the definition is satisfied. ∎

BOTH CONDITIONS, ON THE SAME PAIR

Equal *and* parallel, and it must be the *same* pair of sides. Two sides equal and a different pair parallel gives an isosceles trapezium, not a parallelogram.

Test 2 — both pairs of opposite sides equal

TEST 2

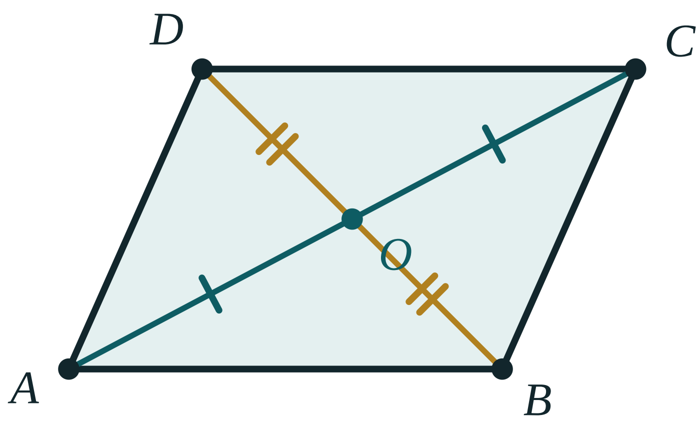
If $AB = CD$ and $BC = AD$, the quadrilateral is a parallelogram.

Proof. The diagonal AC makes $\triangle ABC \cong \triangle CDA$ by SSS. Then both pairs of alternate angles are equal, so both pairs of opposite sides are parallel. ∎

Test 3 — the diagonals bisect each other

TEST 3

If the diagonals of a quadrilateral bisect each other, it is a parallelogram.



$AO = OC$ and $BO = OD$ is enough on its own — the shape has to be a parallelogram.

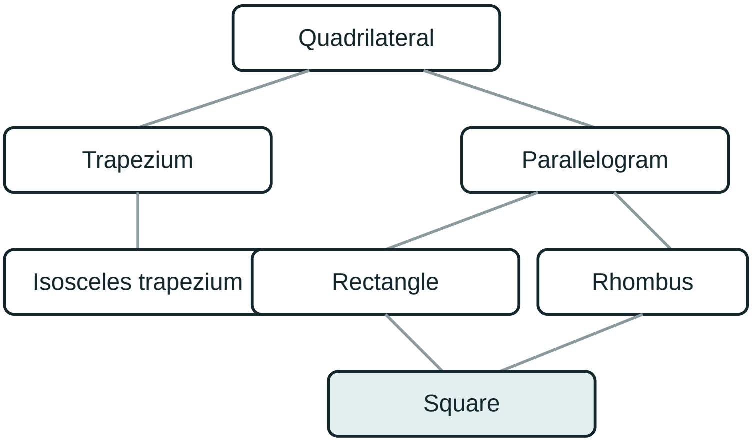
Proof. With $AO = OC$, $BO = OD$ and the vertically opposite angles at O equal, $\triangle AOB \cong \triangle COD$ by SAS. So $AB = CD$, and the equal alternate angles give $AB \parallel CD$. Test 1 finishes the job. ∎

A FOURTH TEST, SOMETIMES USEFUL

If both pairs of opposite **angles** are equal, the quadrilateral is a parallelogram. From $\angle A = \angle C$ and $\angle B = \angle D$ with total $360°$ we get $\angle A + \angle B = 180°$, which makes the sides parallel as co-interior angles.

Conditions that are not enough

- One pair of sides parallel only → a **trapezium**.
- One pair parallel and the *other* pair equal → could be an isosceles trapezium.
- Diagonals equal → could be an isosceles trapezium or a rectangle.
- Two adjacent sides equal → could be a kite.



Where each quadrilateral sits. An arrow means “is a special case of”.

02 Worked examples

EXAMPLE 1 ***In quadrilateral ABCD, AB = 7 cm, CD = 7 cm and AB ∥ CD. Is ABCD a parallelogram?***

1. One pair of opposite sides is both equal and parallel.
   That is exactly Test 1.

2. Test 1 applies to the same pair of sides AB and CD ✓
   Check that the pair really is opposite.

3. So $ABCD$ is a parallelogram.

ANSWER

Yes — by Test 1.

EXAMPLE 2 ***The diagonals of ABCD meet at O with AO = OC = 5 and BO = 4, OD = 6. Is it a parallelogram?***

1. $AO = OC$ ✓

   One diagonal is bisected.

2. $BO = 4 \neq 6 = OD$ ✗

   The other is not.

3. Test 3 needs **both** diagonals bisected.

   One is not enough.

ANSWER

No — Test 3 fails.

EXAMPLE 3 ***M is the midpoint of both AC and BD. Prove ABCD is a parallelogram.***

1. $AM = MC$ and $BM = MD$

   M is the midpoint of each diagonal.

2. So the diagonals bisect each other.

   Exactly the hypothesis of Test 3.

3. $ABCD$ is a parallelogram.

ANSWER

Parallelogram, by Test 3.

03 Interactive model

Use the model to test each condition: which ones force the shape, and which leave it free?

Is this enough to force a parallelogram?

> One pair of opposite sides is equal **and** parallel.

1. Draw diagonal

AC . Alternate angles give

$\angle BAC.$
$=$
$\angle DCA$

Uses
AB
∥
CD.

2. $AB = CD$

, AC
and

is $\triangle ABC \cong \triangle CDA$
common,
so

by
SAS.

3 Then

$\angle BCA$, so
=
$\angle DAC$

$BC \parallel AD$.

Both pairs are now parallel.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Test (criterion)	Alomat	Признак
Sufficient condition	Yetarli shart	Достаточное условие
Necessary condition	Zarur shart	Необходимое условие
Converse statement	Teskari tasdiq	Обратное утверждение
Counter-example	Qarshi misol	Контрпример
Quadrilateral	To‘rtburchak	Четырёхугольник
Trapezium	Trapetsiya	Трапеция
Kite	Deltoid	Дельтоид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which condition forces a quadrilateral to be a parallelogram?

- its diagonals are equal
- its diagonals bisect each other
- two adjacent sides are equal
- one pair of sides is parallel

$AB = CD$ and $AB \parallel CD$ means the quadrilateral is:

- a trapezium
- a parallelogram
- a kite
- not determined

A quadrilateral with one pair of parallel sides only is:

- a parallelogram
- a trapezium
- a rhombus
- a rectangle

Both pairs of opposite angles equal means:

- nothing in particular
- it is a parallelogram
- it is a rectangle
- it is a rhombus

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *$AB = CD$ and $AB \parallel CD$. Is ABCD a parallelogram?*

   ANSWER Yes — Test 1.

2. *$AB = CD$ and $BC = AD$. Is ABCD a parallelogram?*

   ANSWER Yes — Test 2.

3. *The diagonals bisect each other. Is ABCD a parallelogram?*

   ANSWER Yes — Test 3.

4. *Only $AB \parallel CD$ is known. What is ABCD?*

   ANSWER A trapezium.

5. *$\angle A = \angle C$ and $\angle B = \angle D$. Is ABCD a parallelogram?*

   ANSWER Yes.

6. *$AC = BD$. Is ABCD a parallelogram?*

   ANSWER Not necessarily.

7. *$AB = BC = CD = DA$. Is ABCD a parallelogram?*

   ANSWER Yes — a rhombus, which is a parallelogram.

Medium · the standard question

7 PROBLEMS

1. *$AO = OC = 5$, $BO = 4$, $OD = 6$. Is ABCD a parallelogram?*

   ANSWER No — only one diagonal is bisected.

2. *$AB = 7$, $CD = 7$, $BC \parallel AD$. Is ABCD a parallelogram?*

   ANSWER Not necessarily — an isosceles trapezium fits.

3. *M is the midpoint of both AC and BD. Prove ABCD is a parallelogram.*

   ANSWER The diagonals bisect each other — Test 3.

4. *$\angle A = 80°$ and $\angle C = 80°$, $\angle B = 100°$. Find $\angle D$ and decide.*

   ANSWER $\angle D = 100°$; opposite angles equal, so it is a parallelogram.

5. *$AB = 6$, $BC = 9$, $CD = 6$, $DA = 9$. Is ABCD a parallelogram?*

   ANSWER Yes — Test 2.

6. *$AB = 6$, $BC = 9$, $CD = 9$, $DA = 6$. Is ABCD a parallelogram?*

   ANSWER No — the equal sides are adjacent; this is a kite.

7. *Which test proves a rhombus is a parallelogram?*

   ANSWER Test 2 — both pairs of opposite sides are equal.

Hard · stretch and reason 7 PROBLEMS

1. *In $\triangle ABC$, M is the midpoint of BC. Point D is on ray AM with $AM = MD$. Prove $ABDC$ is a parallelogram.*

   ANSWER M is the midpoint of both BC and AD, so the diagonals bisect each other — Test 3.

2. *K and L are the midpoints of AB and CD in parallelogram $ABCD$. Prove $AKCL$ is a parallelogram.*

   ANSWER $AK = \frac{1}{2} AB = \frac{1}{2} CD = CL$ and $AK \parallel CL$ — Test 1.

3. *The diagonals of $ABCD$ meet at O and $\triangle AOB \cong \triangle COD$. Must $ABCD$ be a parallelogram?*

   ANSWER Yes if the congruence pairs AO with CO and BO with DO — then the diagonals bisect each other.

4. *Prove that if the opposite angles of a quadrilateral are equal, it is a parallelogram.*

   ANSWER $2\angle A + 2\angle B = 360°$, so $\angle A + \angle B = 180°$ — co-interior angles make the sides parallel.

5. *Give a quadrilateral with $AB = CD$, $AD = BC$ that is not a parallelogram, or explain why none exists.*

   ANSWER None exists — Test 2 proves every such quadrilateral is a parallelogram (for a simple, non-crossed quadrilateral).

6. *E is the midpoint of AB and F of DC in parallelogram $ABCD$. Prove $AEFD$ is a parallelogram.*

   ANSWER $AE = \frac{1}{2} AB = \frac{1}{2} DC = DF$ and $AE \parallel DF$ — Test 1.

7. *In parallelogram $ABCD$, points E and F lie on the diagonal AC with $AE = FC$. Prove $BEDF$ is a parallelogram.*

   ANSWER The diagonals BD and EF share the midpoint O of AC, so they bisect each other — Test 3.

07 Homework

Homework — 5 problems

Geometry 8, Тема 3, pp. 11–13. Name the test you used in each proof.

1. *$AB = 9$, $CD = 9$ and $AB \parallel CD$. Prove $ABCD$ is a parallelogram and name the test.*
2. *$AO = 4$, $OC = 4$, $BO = 7$, $OD = 7$. Prove $ABCD$ is a parallelogram.*
3. *Give an example of a quadrilateral with equal diagonals that is not a parallelogram.*
4. *In $\triangle ABC$, M is the midpoint of AC and BM is extended to D with $BM = MD$. Prove $ABCD$ is a parallelogram.*
5. Write the three tests in your own words, each with a small labelled sketch.

Rectangle and its properties

A parallelogram with one right angle — which turns out to force all four, and makes the diagonals equal.

LESSON 7 | 1 × 40 MIN | GEOMETRY 8, TEMA 4 | STAGE 9 · BEYOND

LESSON CLOCK

4' | 12' | 8' | 14' | 2'

Warm-up	4 min
Explanation	12 min
Interactive	8 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Define a rectangle and show that one right angle forces four.
- Prove that the diagonals of a rectangle are equal.
- Use the fact that the diagonals bisect each other and are equal.
- Apply Pythagoras to find a diagonal from the sides.

TEXTBOOK

National	Тема 4, pp. 14–15
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Definition

DEFINITION

A **rectangle** is a parallelogram with a right angle.

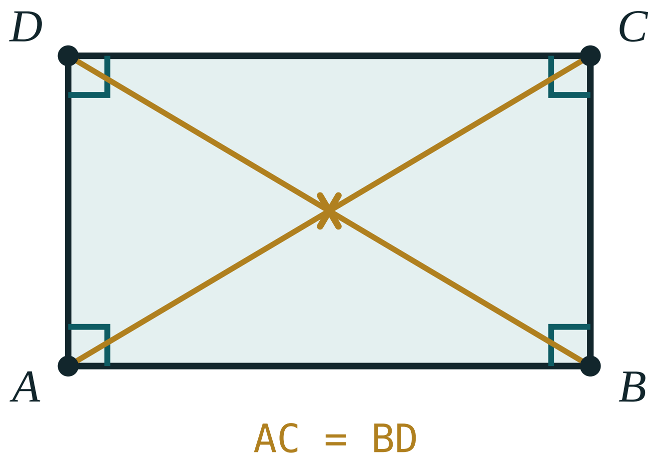
One right angle is enough. Neighbouring angles of a parallelogram are supplementary, so if $\angle A = 90°$ then $\angle B = 90°$, and opposite angles give $\angle C = \angle D = 90°$.

Because a rectangle *is* a parallelogram, it inherits everything from Тема 2: opposite sides equal, opposite angles equal, diagonals bisecting each other. Only the new property has to be proved.

The new property — equal diagonals

THEOREM

The diagonals of a rectangle are equal: $AC = BD$.



Four right angles, and two diagonals of exactly the same length crossing at the centre.

Proof. Compare $\triangle ABC$ and $\triangle BAD$:

- AB is common;
- $BC = AD$ — opposite sides of a parallelogram;
- $\angle ABC = \angle BAD = 90°$.

So $\triangle ABC \cong \triangle BAD$ by SAS, and therefore $AC = BD$. ∎

CONSEQUENCE

The diagonals bisect each other **and** are equal, so all four half-diagonals are equal: $AO = BO = CO = DO$. The point O is the same distance from all four vertices — it is the centre of the circle through them.

The test

TEST

A parallelogram whose diagonals are equal is a rectangle.

Note carefully: *a parallelogram* whose diagonals are equal. Without that word the statement is false — an isosceles trapezium also has equal diagonals.

The diagonal from the sides

A diagonal cuts the rectangle into two right-angled triangles, so Pythagoras applies:

$$d = \sqrt{a^2 + b^2}$$

A 6×8 rectangle has diagonal $\sqrt{36 + 64} = \sqrt{100} = 10$.

And the other two familiar formulae: perimeter $P = 2(a + b)$, area $S = ab$.

02 Worked examples

EXAMPLE 1 **A rectangle has sides 9 cm and 12 cm. Find its diagonal, perimeter and area.**

1. $d = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$

   Pythagoras on half the rectangle.

2. $d = 15\ cm$

3. $P = 2(9 + 12) = 42\ cm$

4. $S = 9 \cdot 12 = 108\ cm^2$

ANSWER

$d = 15\ cm,\ P = 42\ cm,\ S = 108\ cm^2$

EXAMPLE 2 ***In rectangle ABCD the diagonals meet at O and ∠AOB = 60°. Find ∠ACB.***

1. $AO = BO$

   Half-diagonals of a rectangle are all equal.

2. $\triangle AOB$ is isosceles with apex $60°$, so it is equilateral.

   All three angles are $60°$.

3. $\angle OAB = 60°$

   That is $\angle CAB$.

4. $\angle ACB = 90° - 60° = 30°$

   The acute angles of right triangle ABC add to $90°$.

ANSWER

$\angle ACB = 30°$

EXAMPLE 3 **The diagonal of a rectangle is 13 cm and one side is 5 cm. Find the area.**

1. $b = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144}$

   Pythagoras, rearranged.

2. $b = 12\ cm$

3. $S = 5 \cdot 12 = 60\ cm^2$

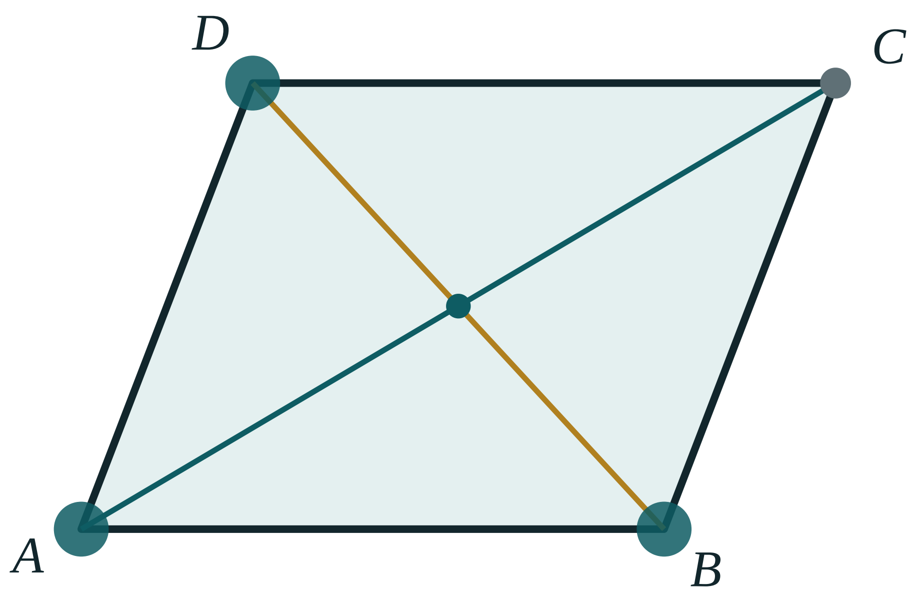
ANSWER

$60\ cm^2$

03 Interactive model

Drag until ∠A reads 90° and watch AC and BD become equal — then drag away and watch them separate again.

When do the diagonals become equal?



∠A **69°**	AC **255.5**	BD **176.9**	AO **127.8**	OC **127.8**	AB **170**	AD **139.3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rectangle	To‘g‘ri to‘rtburchak	Прямоугольник
Right angle	To‘g‘ri burchak	Прямой угол
Equal diagonals	Teng diagonallar	Равные диагонали
Area	Yuza	Площадь
Pythagoras’ theorem	Pifagor teoremasi	Теорема Пифагора
Hypotenuse	Gipotenuza	Гипотенуза
Circumscribed circle	Tashqi chizilgan aylana	Описанная окружность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rectangle is defined as:

- a quadrilateral with a right angle
- a parallelogram with a right angle
- a parallelogram with equal sides
- a quadrilateral with equal diagonals

In a rectangle the diagonals are:

- perpendicular
- equal and bisect each other
- equal but do not bisect
- angle bisectors

A rectangle has sides 8 and 15. Its diagonal is:

- 17
- 23
- $\sqrt{23}$
- 120

A quadrilateral with equal diagonals must be a rectangle:

- true
- false — it could be an isosceles trapezium
- true only if it is convex
- true only if the sides are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rectangle has sides 3 cm and 4 cm. Find its diagonal.*

   ANSWER *5 cm*

2. *Sides 6 cm and 8 cm. Find the diagonal.*

   ANSWER *10 cm*

3. *Sides 5 cm and 7 cm. Find the perimeter.*

   ANSWER *24 cm*

4. *Sides 5 cm and 7 cm. Find the area.*

   ANSWER *35 cm^2*

5. *$AC = 10$ cm in rectangle ABCD. Find BD.*

   ANSWER *10 cm*

6. *$AC = 10$ cm and the diagonals meet at O. Find BO.*

   ANSWER *5 cm*

7. *One angle of a rectangle is 90°. Find the other three.*

   ANSWER *90°* each

Medium · the standard question

7 PROBLEMS

1. *Sides 9 cm and 12 cm. Find the diagonal, perimeter and area.*

   ANSWER *15 cm, 42 cm, 108 cm²*

2. *The diagonal is 13 cm and one side is 5 cm. Find the area.*

   ANSWER *60 cm²*

3. *The perimeter is 34 cm and one side is 5 cm. Find the diagonal.*

   ANSWER other side *12 cm*, diagonal *13 cm*

4. *The diagonals meet at O and $\angle AOB = 60°$. Find $\angle ACB$.*

   ANSWER *30°*

5. *The area is 48 cm² and one side is 6 cm. Find the perimeter.*

   ANSWER *28 cm*

6. *The diagonals of a rectangle are 20 cm. Find the distance from O to each vertex.*

   ANSWER *10 cm*

7. *A rectangle has a diagonal of 25 cm and a side of 20 cm. Find the perimeter.*

   ANSWER other side *15 cm*, *$P = 70$ cm*

Hard · stretch and reason

7 PROBLEMS

1. *The diagonals meet at O and $\angle AOD = 120°$. Find the acute angle between a diagonal and the longer side.*

   ANSWER $\angle AOB = 60°$, so $\triangle AOB$ is equilateral and the angle is $60°$.

2. *A rectangle has perimeter 46 cm and diagonal 17 cm. Find its sides.*

   ANSWER $a + b = 23$, $a^2 + b^2 = 289 \Longrightarrow ab = 120 \Longrightarrow$ sides $8\ cm$ and $15\ cm$

3. *Prove that a parallelogram with equal diagonals is a rectangle.*

   ANSWER The diagonals bisect each other, so $\triangle ABC \cong \triangle BAD$ by SSS; then $\angle A = \angle B$ and they are supplementary, so each is $90°$.

4. *In rectangle ABCD the bisector of $\angle A$ meets BC at K, with BK = 4 cm and KC = 6 cm. Find the perimeter.*

   ANSWER $\triangle ABK$ is an isosceles right triangle, so $AB = BK = 4$ and $BC = 10$: $P = 28\ cm$.

5. *A rectangle is inscribed in a circle of radius 6.5 cm and one side is 5 cm. Find the other side.*

   ANSWER The diagonal is the diameter, $13\ cm$; the other side is $12\ cm$.

6. *Show that the point where the diagonals of a rectangle meet is equidistant from all four vertices.*

   ANSWER The diagonals bisect each other and are equal, so $AO = BO = CO = DO = \frac{d}{2}$.

7. *A rectangle has area 120 cm² and diagonal 17 cm. Find its sides.*

   ANSWER $ab = 120$ and $a^2 + b^2 = 289$, so $(a + b)^2 = 529$ and $a + b = 23$: the sides are $8\ cm$ and $15\ cm$.

07 Homework

Homework — 5 problems

Geometry 8, Тема 4, pp. 14–15.

1. *A rectangle has sides 7 cm and 24 cm. Find the diagonal, perimeter and area.*

2. *The diagonal is 20 cm and one side is 16 cm. Find the other side and the area.*
3. *The diagonals of rectangle ABCD meet at O and ∠BOC = 120°. Find ∠OBC.*
4. *The perimeter of a rectangle is 40 cm and one side is 4 cm longer than the other. Find the diagonal.*
5. *Prove that the diagonals of a rectangle are equal, stating the congruence test you use.*

Rhombus and square

A parallelogram with equal sides, and the shape that is a rectangle and a rhombus at the same time.

LESSON 8–9 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 5–6 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 13' | 8' | 12' | 2'

Warm-up	5 min
Explanation	13 min
Interactive	8 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Define a rhombus and prove that its diagonals are perpendicular and bisect its angles.
- Define a square and list the properties it inherits from both parents.
- Use the area formula $S = 1⁄2 d_1 d_2$ for a rhombus.
- Place every quadrilateral of Chapter I correctly in the family tree.

TEXTBOOK

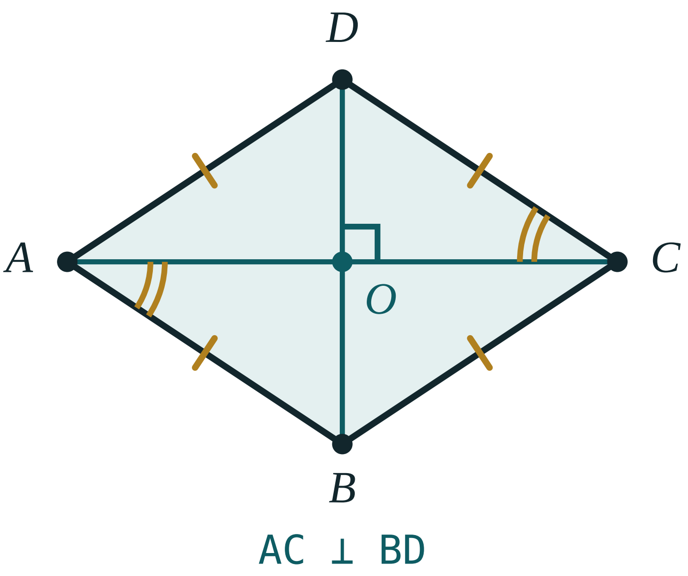
National	Темы 5–6, pp. 16–18
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The rhombus

DEFINITION

A **rhombus** is a parallelogram all of whose sides are equal.



All four sides equal. The diagonals meet at right angles and each one bisects the two angles it passes through.

It inherits everything a parallelogram has, and gains two properties of its own.

> THEOREM
>
> The diagonals of a rhombus are **perpendicular**, and each diagonal **bisects** the two angles through which it passes.

Proof. $\triangle ABC$ has $AB = BC$, so it is isosceles. The diagonal BD passes through O, the midpoint of AC (parallelogram property), so BO is the median of this isosceles triangle from its apex — and therefore also its height and its angle bisector. That gives $BD \perp AC$ and $\angle ABD = \angle DBC$ at once. ∎

The area of a rhombus

The perpendicular diagonals cut the rhombus into four congruent right triangles with legs $\frac{d_1}{2}$ and $\frac{d_2}{2}$. Adding their areas:

$$S = 4 \cdot \frac{1}{2} \cdot \frac{d_1}{2} \cdot \frac{d_2}{2} = \frac{1}{2} d_1 d_2$$

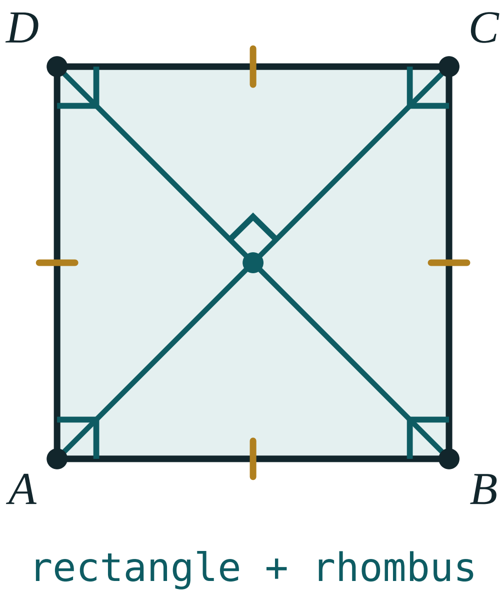
And Pythagoras in one of those triangles relates the side to the diagonals:

$$a^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

The square

DEFINITION

A **square** is a rectangle all of whose sides are equal — equivalently, a rhombus with a right angle.



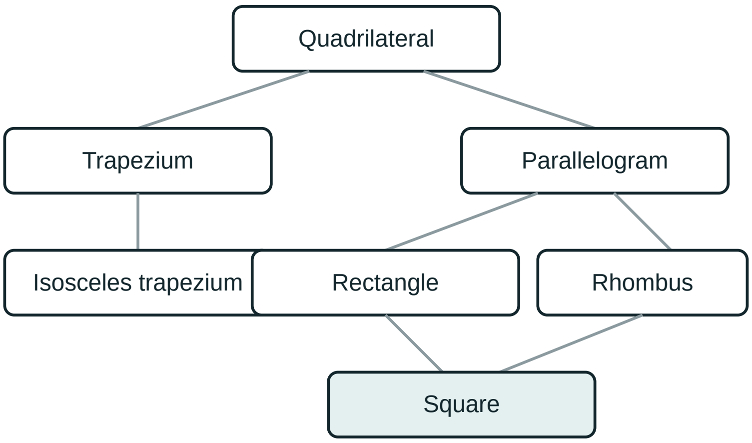
A square has every property of a rectangle and every property of a rhombus at once.

So a square has **all** of the following:

- four equal sides and four right angles;
- diagonals that bisect each other (parallelogram);
- diagonals that are **equal** (rectangle);
- diagonals that are **perpendicular** and bisect the angles into $45°$ halves (rhombus).

$$d = a\sqrt{2} \quad \cdot \quad S = a^2 = \frac{d^2}{2}$$

The family tree



Each arrow means "is a special case of". The square sits at the bottom because it inherits from both the rectangle and the rhombus.

Read it downwards and every statement is true: a square *is* a rectangle, a rectangle *is* a parallelogram, a parallelogram *is* a quadrilateral. Read it upwards and almost every statement is false — that asymmetry is the point of the diagram.

02 Worked examples

EXAMPLE 1 **The diagonals of a rhombus are 12 cm and 16 cm. Find its side, perimeter and area.**

1. *half-diagonals: 6 cm and 8 cm*

   The diagonals bisect each other.

2. $a = \sqrt{6^2 + 8^2} = \sqrt{100} = 10\ cm$

   Pythagoras in one of the four right triangles.

3. $P = 4 \cdot 10 = 40\ cm$

   All four sides are equal.

4. $S = \frac{1}{2} \cdot 12 \cdot 16 = 96\ cm^2$

   Half the product of the diagonals.

ANSWER

$a = 10\ cm,\ P = 40\ cm,\ S = 96\ cm^2$

EXAMPLE 2

In rhombus ABCD, $\angle A = 60°$. Find $\angle ABD$ and prove that the shorter diagonal equals the side.

1. $\angle B = 180° - 60° = 120°$

   Neighbouring angles of a parallelogram.

2. $\angle ABD = \dfrac{120°}{2} = 60°$

   The diagonal bisects the angle.

3. $\triangle ABD$ has $\angle A = 60°$ and $\angle ABD = 60°$

   so the third angle is $60°$ too.

4. $\triangle ABD$ is equilateral, so $BD = AB$

   The shorter diagonal equals the side.

ANSWER

$\angle ABD = 60°$; $BD = AB$

EXAMPLE 3 **A square has diagonal 8 cm. Find its side and area.**

1. $d = a\sqrt{2}$

   The diagonal of a square.

2. $a = \dfrac{8}{\sqrt{2}} = 4\sqrt{2}\ cm$

   Rationalise the denominator.

3. $S = \dfrac{d^2}{2} = \dfrac{64}{2} = 32\ cm^2$

   Quicker than squaring the side.

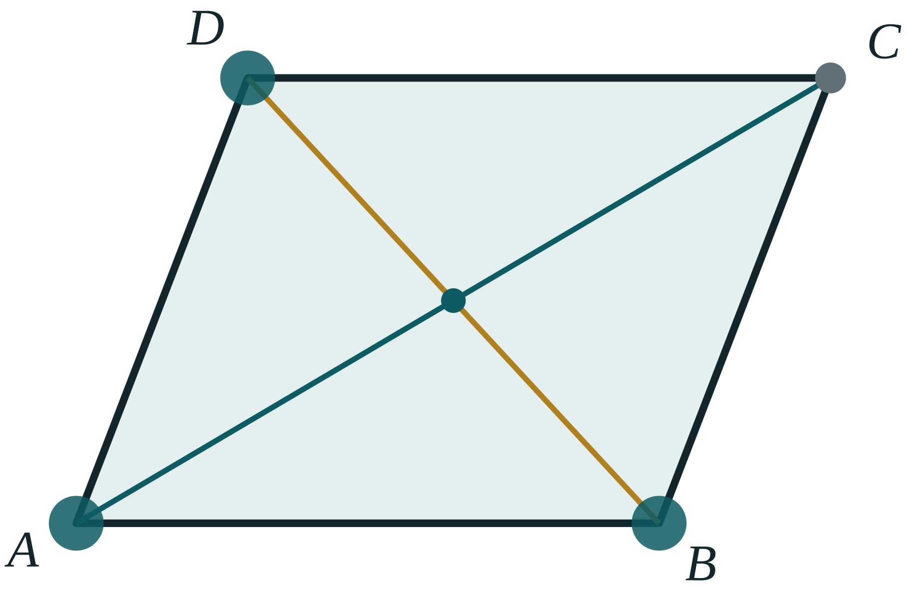
ANSWER

$a = 4\sqrt{2}\ cm \approx 5.66\ cm,\ S = 32\ cm^2$

03 Interactive model

Switch the model to rhombus mode: all four sides stay equal however you drag, and AC ⊥ BD holds throughout.

Drag the rhombus



AB 170	BC 139.3	DC 170	AD 139.3	AC 255.5	BD 176.9	AO 127.8
OC 127.8						

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rhombus	Romb	Ромб
Square	Kvadrat	Квадрат
Perpendicular	Perpendikulyar	Перпендикулярный
Bisects the angle	Burchakni teng ikkiga bo‘ladi	Делит угол пополам
Half-diagonal	Diagonalning yarmi	Половина диагонали
Equal sides	Teng tomonlar	Равные стороны
Classification	Tasnif	Классификация

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The diagonals of a rhombus are:

- equal
- perpendicular
- equal and perpendicular
- neither

A rhombus with diagonals 6 and 8 has area:

- *48*
- *24*
- *14*
- *10*

A square is:

- a rectangle only
- a rhombus only
- both a rectangle and a rhombus
- neither

The diagonal of a square of side 5 is:

- 10
- $5\sqrt{2}$
- $\sqrt{5}$
- 25

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rhombus has side 7 cm. Find its perimeter.*

   ANSWER *28 cm*

2. *The diagonals of a rhombus are 6 cm and 8 cm. Find its area.*

   ANSWER *24 cm²*

3. *In a rhombus AC = 10 cm. Find AO.*

   ANSWER *5 cm*

4. *One angle of a rhombus is 70°. Find the neighbouring angle.*

   ANSWER *110°*

5. *A square has side 6 cm. Find its perimeter and area.*

   ANSWER *24 cm, 36 cm²*

6. *A square has side 5 cm. Find its diagonal.*

   ANSWER $5\sqrt{2}$ *cm*

7. *What is the angle between the diagonals of a rhombus?*

   ANSWER *90°*

Medium · the standard question

7 PROBLEMS

1. *The diagonals of a rhombus are 12 cm and 16 cm. Find its side and perimeter.*

   ANSWER *10 cm, 40 cm*

2. *A rhombus has side 13 cm and one diagonal 10 cm. Find the other diagonal.*

   ANSWER *24 cm*

3. *In rhombus ABCD, $\angle A = 60°$. Find $\angle ABD$.*

   ANSWER *60°*

4. *A square has diagonal 8 cm. Find its side and area.*

   ANSWER $4\sqrt{2}$ *cm, 32 cm²*

5. *A rhombus has perimeter 40 cm and one diagonal 12 cm. Find the other.*

   ANSWER *16 cm*

6. *The area of a rhombus is 60 cm² and one diagonal is 10 cm. Find the other.*

   ANSWER *12 cm*

7. *A square has area 49 cm². Find its diagonal.*

   ANSWER $7\sqrt{2}$ *cm*

Hard · stretch and reason

7 PROBLEMS

1. *In rhombus ABCD, $\angle A = 60°$. Prove that the shorter diagonal equals the side.*

   ANSWER $\triangle ABD$ has three $60°$ angles, so it is equilateral and $BD = AB$.

2. *A rhombus has side 10 cm and height 8 cm. Find its area, then its diagonals if one is 16 cm.*

   ANSWER $S = 10 \cdot 8 = 80$; then $\frac{1}{2} \cdot 16 \cdot d_2 = 80$ gives $d_2 = 10$ cm.

3. *Prove that a parallelogram whose diagonals are perpendicular is a rhombus.*

   ANSWER The diagonals bisect each other, so each half-diagonal pair gives congruent right triangles by SAS; all four sides come out equal.

4. *The diagonals of a square are 14 cm. Find its side, perimeter and area.*

   ANSWER $a = 7\sqrt{2}$, $P = 28\sqrt{2}$ cm, $S = 98$ cm²

5. *A rhombus has one angle of 120° and side 6 cm. Find both diagonals.*

   ANSWER Shorter $= 6$ cm (equilateral half), longer $= 6\sqrt{3}$ cm

6. *Prove that the midpoints of the sides of a rectangle form a rhombus.*

   ANSWER Each side of the new quadrilateral is a midline equal to half a diagonal, and the diagonals of a rectangle are equal — so all four sides are equal.

7. *A square and a rhombus both have side 8 cm. Compare their perimeters, and explain why their areas can differ.*

   ANSWER Both perimeters are 32 cm. The rhombus can be flattened, which reduces its height and so its area; the square is the tallest case.

07 Homework

Homework — 6 problems

Geometry 8, Темы 5–6, pp. 16–18.

1. *The diagonals of a rhombus are 10 cm and 24 cm. Find its side, perimeter and area.*
2. *A rhombus has side 17 cm and one diagonal 16 cm. Find the other diagonal.*

3. *In rhombus ABCD, ∠B = 120°. Find ∠BAC.*
4. *A square has perimeter 32 cm. Find its diagonal and area.*
5. *A square has diagonal 10 cm. Find its area.*
6. *Draw the quadrilateral family tree from memory and mark where the rhombus and the square sit.*

Home / Grade 8 / Geometry / Quarter 1

Trapezium and the isosceles trapezium

Exactly one pair of parallel sides — and what happens when the two legs are made equal.

LESSON 10–11 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 7–8 | STAGE 9 · BEYOND

LESSON CLOCK

Warm-up	5 min
Explanation	13 min
Interactive	8 min
Practice	12 min
Homework	2 min

LEARNING OBJECTIVES

- Name the parts of a trapezium: bases, legs, height.
- Use the fact that the angles on each leg are supplementary.
- Prove and use the properties of an isosceles trapezium.
- Recognise a right-angled trapezium and solve problems using its height.

TEXTBOOK

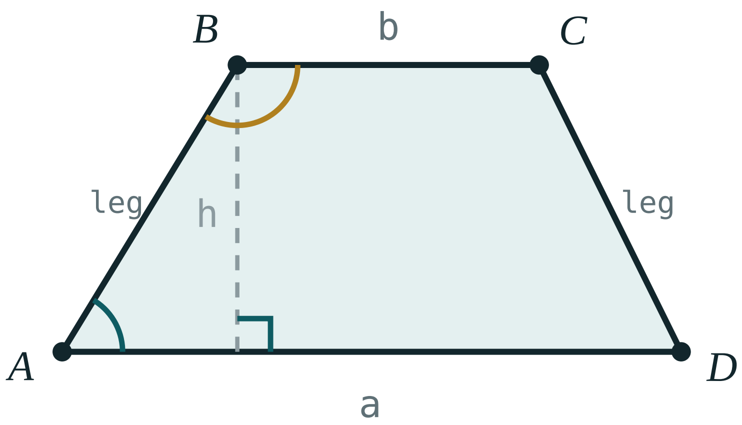
National	Темы 7–8, pp. 19–22
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Definition and parts

DEFINITION

A **trapezium** is a quadrilateral with exactly one pair of parallel sides.



The parallel sides a and b are the bases; the other two are the legs; h is the distance between the bases.

- The parallel sides are the **bases** — the longer one a, the shorter b.
- The other two sides are the **legs**.
- The **height** h is the distance between the two bases, measured perpendicular to them.

ANGLES ON A LEG

The two angles at the ends of a leg are co-interior angles between the parallel bases, so

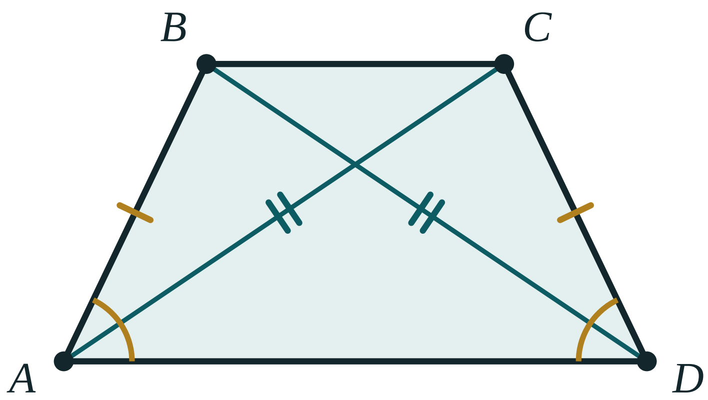
$$\angle A + \angle B = 180° \quad \text{and} \quad \angle C + \angle D = 180°$$

A trapezium with a right angle has two of them (a **right-angled trapezium**), because the angle next to it on the same leg must also be $90°$.

The isosceles trapezium

DEFINITION

A trapezium whose two legs are equal is **isosceles**.



AB = CD · ∠A = ∠D · AC = BD

Equal legs, equal base angles and equal diagonals — the three properties travel together.

THEOREM

In an isosceles trapezium:

1. the angles at each base are equal: $\angle A = \angle D$ and $\angle B = \angle C$;
2. the diagonals are equal: $AC = BD$.

Proof of (1). Drop perpendiculars from B and C onto AD. The two right triangles formed have equal hypotenuses (the legs) and equal heights, so they are congruent by RHS — giving $\angle A = \angle D$. The other pair follows from the supplementary angles on each leg.

Proof of (2). $\triangle ABD$ and $\triangle DCA$ have AD common, $AB = DC$, and the included angles $\angle A = \angle D$ — congruent by SAS, so $BD = AC$. ∎

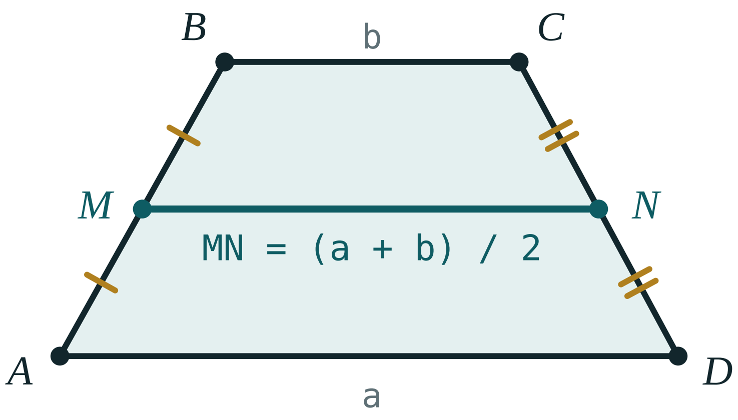
EQUAL DIAGONALS ARE NOT ENOUGH ON THEIR OWN

A rectangle also has equal diagonals. The statement “a trapezium with equal diagonals is isosceles” is true, but the word *trapezium* is doing real work.

The midline (a first look)

The segment joining the midpoints of the two legs is the **midline**. It is parallel to the bases and its length is their average:

$$MN = \frac{a + b}{2}$$



The midline joins the midpoints of the legs and measures exactly the average of the two bases.

The full proof comes in the next lesson but two; for now, use the formula and check it against the interactive model.

Working with the height

Almost every numerical trapezium problem is solved by dropping one or two perpendiculars from the shorter base to the longer one. That cuts the trapezium into a rectangle and one or two right triangles, and then Pythagoras does the rest.

For an **isosceles** trapezium the two triangles are congruent, so each cuts off $\frac{a-b}{2}$ from the longer base — a formula worth remembering.

$$S = \frac{a+b}{2} \cdot h = MN \cdot h$$

02 Worked examples

EXAMPLE 1 ***In trapezium ABCD with AD ∥ BC, ∠A = 65°. Find ∠B.***

1. AB is a leg joining the two parallel bases.

2. $\angle A + \angle B = 180°$

   Co-interior angles.

3. $\angle B = 115°$

ANSWER

$\angle B = 115°$

EXAMPLE 2 **An isosceles trapezium has bases 10 cm and 4 cm and legs 5 cm. Find its height and area.**

1. $\frac{a - b}{2} = \frac{10 - 4}{2} = 3\ cm$

   Each right triangle cuts off 3 cm.

2. $h = \sqrt{5^2 - 3^2} = \sqrt{16} = 4\ cm$

   Pythagoras in one triangle.

3. $MN = \frac{10 + 4}{2} = 7\ cm$

   The midline.

4. $S = 7 \cdot 4 = 28\ cm^2$

ANSWER

$h = 4\ cm,\ S = 28\ cm^2$

EXAMPLE 3 **The midline of a trapezium is 9 cm and one base is 5 cm. Find the other base.**

1. $\frac{a + b}{2} = 9$

   The midline formula.

2. $a + b = 18$

3. $a = 18 - 5 = 13$ cm

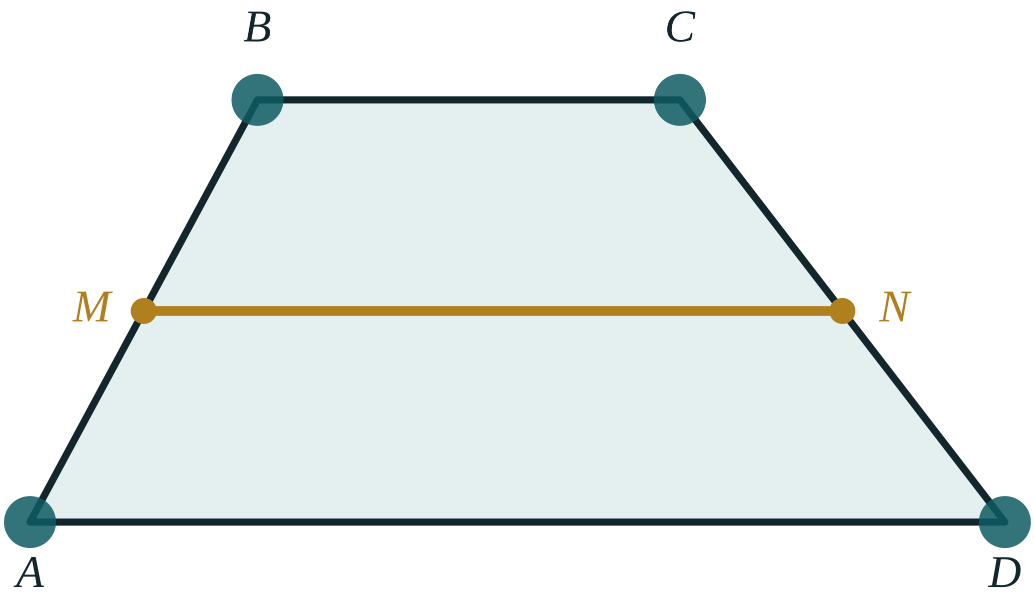
ANSWER

13 cm

03 Interactive model

Drag the corners and read the three numbers: the midline always sits exactly between the two bases.

The midline of a trapezium



a (AD) **300** | b (BC) **130** | (a + b) / 2 **215** | MN measured **215**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Trapezium	Trapetsiya	Трапеция
Base of a trapezium	Trapetsiya asosi	Основание трапеции
Leg (lateral side)	Yon tomon	Боковая сторона
Height	Balandlik	Высота
Isosceles trapezium	Teng yonli trapetsiya	Равнобедренная трапеция
Right-angled trapezium	To‘g‘ri burchakli trapetsiya	Прямоугольная трапеция
Midline	O‘rta chiziq	Средняя линия
Half-sum	Yarim yig‘indi	Полусумма

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A trapezium has:

- both pairs of sides parallel
- exactly one pair of parallel sides
- four equal sides
- perpendicular diagonals

In trapezium ABCD with AD ∥ BC, $\angle A = 72°$. Then $\angle B$ is:

- $72°$
- $108°$
- $18°$
- not determined

The diagonals of an isosceles trapezium are:

- perpendicular
- equal
- bisecting each other
- angle bisectors

The midline of a trapezium with bases 6 and 10 is:

- 4
- 8
- 16
- 60

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *In trapezium ABCD with $AD \parallel BC$, $\angle A = 65°$. Find $\angle B$.*

   ANSWER *115°*

2. *$\angle C = 105°$. Find $\angle D$.*

   ANSWER *75°*

3. *The bases are 8 cm and 12 cm. Find the midline.*

   ANSWER *10 cm*

4. *The bases are 5 cm and 9 cm and the height is 4 cm. Find the area.*

   ANSWER *28 cm²*

5. *An isosceles trapezium has $\angle A = 70°$. Find $\angle D$.*

   ANSWER *70°*

6. *An isosceles trapezium has AC = 13 cm. Find BD.*

   ANSWER *13 cm*

7. *A right-angled trapezium has one angle of 90°. How many right angles does it have?*

   ANSWER *2*

Medium · the standard question

7 PROBLEMS

1. *An isosceles trapezium has bases 10 cm and 4 cm and legs 5 cm. Find the height.*

   ANSWER *4 cm*

2. *The same trapezium: find its area.*

   ANSWER *28 cm²*

3. *The midline is 9 cm and one base is 5 cm. Find the other base.*

   ANSWER *13 cm*

4. *An isosceles trapezium has* $\angle A = 60°$*, bases 12 cm and 6 cm. Find the leg.*

   ANSWER $\frac{12 - 6}{2} = 3$, so the leg is 6 *cm*

5. *The area is 48 cm² and the height is 6 cm. Find the midline.*

   ANSWER *8 cm*

6. *An isosceles trapezium has perimeter 34 cm, bases 6 cm and 14 cm. Find each leg.*

   ANSWER *7 cm*

7. *In a right-angled trapezium the bases are 9 cm and 5 cm and the slant leg is 5 cm. Find the height.*

   ANSWER $h = \sqrt{5^2 - 4^2} = 3$ *cm*

Hard · stretch and reason

7 PROBLEMS

1. *Prove that a trapezium with equal base angles is isosceles.*

ANSWER Drop the two heights: the right triangles are congruent by ASA, so the legs are equal.

2. *An isosceles trapezium has diagonals that are perpendicular, bases 6 cm and 10 cm. Find its height and area.*

ANSWER Perpendicular diagonals give $h = \frac{a+b}{2} = 8\ cm$ and $S = 8 \cdot 8 = 64\ cm^2$.

3. *In trapezium ABCD, AD ∥ BC, BC = 6 cm, and the diagonal AC bisects ∠A. Prove that AB = BC and find AB.*

ANSWER $\angle CAD = \angle ACB$ (alternate angles) and $\angle CAD = \angle CAB$ (bisector), so $\angle CAB = \angle ACB$ and $\triangle ABC$ is isosceles: $AB = BC = 6\ cm$.

4. *An isosceles trapezium has bases 9 cm and 21 cm and legs 10 cm. Find its height and area.*

ANSWER $\frac{21-9}{2} = 6$, so $h = 8\ cm$ and $S = 120\ cm^2$.

5. *Prove that the diagonals of an isosceles trapezium are equal.*

ANSWER $\triangle ABD \cong \triangle DCA$ by SAS: AD common, $AB = DC$, $\angle A = \angle D$.

6. *An isosceles trapezium has perimeter 48 cm, bases 10 cm and 18 cm. Find each leg and the height.*

ANSWER legs $10\ cm$; $\frac{18-10}{2} = 4$, so $h = \sqrt{100-16} = 2\sqrt{21} \approx 9.2\ cm$.

7. *Show that the height of an isosceles trapezium with bases a, b and leg c is* $\sqrt{c^2 - \left(\frac{a-b}{2}\right)^2}$

ANSWER Each right triangle has hypotenuse c and horizontal leg $\frac{a-b}{2}$; Pythagoras gives the height.

07 Homework

Homework — 6 problems

Geometry 8, Темы 7–8, pp. 19–22. Draw the height on every figure.

1. *In trapezium ABCD with $AD \parallel BC$, $\angle B = 128°$. Find $\angle A$.*
2. *The bases are 7 cm and 13 cm. Find the midline.*
3. *An isosceles trapezium has bases 16 cm and 6 cm and legs 13 cm. Find its height and area.*
4. *The midline is 11 cm and one base is 8 cm. Find the other.*
5. *An isosceles trapezium has perimeter 44 cm and bases 8 cm and 16 cm. Find each leg.*
6. *Prove that the base angles of an isosceles trapezium are equal.*

Thales’ theorem

Parallel lines cutting equal pieces from one transversal cut equal pieces from every other transversal.

LESSON 12 | 1 × 40 MIN | GEOMETRY 8, TEMA 9 | STAGE 9 · 13.2, 13.4

LESSON CLOCK

4' | 13' | 8' | 13' | 2'

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State Thales’ theorem precisely.
- Use it to divide a segment into a given number of equal parts.
- Use the proportional form to find an unknown length.
- Recognise when the theorem does *not* apply.

TEXTBOOK

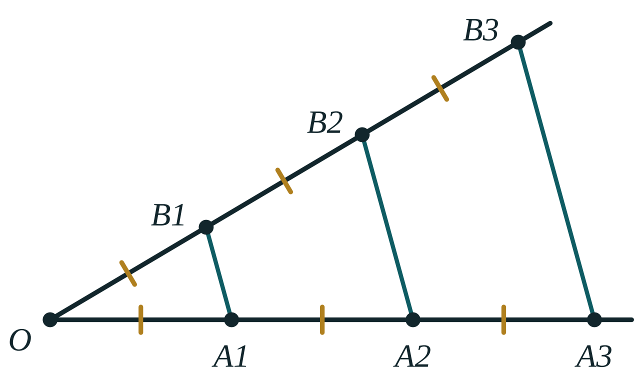
National	Тема 9, pp. 23–25
Cambridge	Learner’s Book pp. 278–299
Workbook	Workbook 13.2

01 Explanation

The theorem

THALES’ THEOREM

If parallel lines cut equal segments on one transversal, then they cut equal segments on every other transversal.



Three parallel lines cut equal segments on the lower ray, and therefore cut equal segments on the upper ray as well.

In the figure, $OA_1 = A_1A_2 = A_2A_3$ on the lower ray forces $OB_1 = B_1B_2 = B_2B_3$ on the upper one — even though the two rays make different angles and the segments have different lengths.

THE PROPORTIONAL FORM

More generally, parallel lines cut **proportional** segments:

$$\frac{\mathrm{OA}_1}{\mathrm{OA}_2} = \frac{\mathrm{OB}_1}{\mathrm{OB}_2} \quad \text{and} \quad \frac{\mathrm{OA}_1}{\mathrm{A}_1\mathrm{A}_2} = \frac{\mathrm{OB}_1}{\mathrm{B}_1\mathrm{B}_2}$$

This is the form you will use for calculations; the equal-segments version is the special case where the ratio is 1.

Why it is true

Through B_1 draw a line parallel to the lower ray; it cuts the next parallel at a point P. Then OA_1B_1 and B_1PB_2 are triangles with equal corresponding angles (alternate and corresponding angles on the parallels) and one equal side ($A_1A_2 = B_1P$, opposite sides of a parallelogram). They are congruent by ASA, so $OB_1 = B_1B_2$. Repeating the argument along the ray finishes the proof. ∎

Dividing a segment into equal parts

This is the classic construction, and it needs no measurement at all.

1. To divide AB into 5 equal parts, draw any ray from A.
2. With compasses, step off 5 equal segments along the ray, ending at P_5.
3. Join P_5 to B.
4. Through P_1, P_2, P_3, P_4 draw lines parallel to P_5B.
5. By Thales' theorem these lines cut AB into 5 equal parts.

WHY THIS MATTERS

Bisecting a segment is easy with compasses. Cutting it into **five** equal parts is not — and Thales' theorem does it exactly, with no arithmetic and no ruler markings.

When it does not apply

THE LINES MUST BE PARALLEL

> Three lines that merely look evenly spread do not give equal cuts. If the figure does not say “parallel” — or you cannot prove it — the theorem is not available.

And the equal segments must be measured *along the transversals*, not along the parallel lines themselves.

02 Worked examples

EXAMPLE 1 ***Parallel lines cut a transversal so that $OA_1 = A_1A_2 = A_2A_3 = 4$ cm, and on a second transversal $OB_1 = 3$ cm. Find OB_3.***

1. $OB_1 = B_1B_2 = B_2B_3$

   Thales’ theorem — equal cuts on one transversal give equal cuts on the other.

2. $B_1B_2 = B_2B_3 = 3\ cm$

3. $OB_3 = 3 \cdot 3 = 9\ cm$

ANSWER

$OB_3 = 9\ cm$

EXAMPLE 2 ***In the proportional form, $OA_1 = 6$, $OA_2 = 15$ and $OB_1 = 8$. Find OB_2.***

1. $$\frac{OA_1}{OA_2} = \frac{OB_1}{OB_2}$$

   Set up the proportion.

2. $$\frac{6}{15} = \frac{8}{OB_2}$$

   Substitute.

3. $6 \cdot OB_2 = 15 \cdot 8 = 120$

   Cross-multiply.

4. $OB_2 = 20$

ANSWER

$OB_2 = 20$

EXAMPLE 3 **Describe how to divide a segment of length 7 cm into 3 equal parts without measuring.**

1. Draw a ray from one end A at any convenient angle.

   The angle does not matter.

2. Step off three equal compass widths along the ray: P_1, P_2, P_3.

   Any width will do.

3. Join P_3 to the other end B.

4. Through P_1 and P_2 draw lines parallel to P_3B.

   They cut AB into three equal parts by Thales.

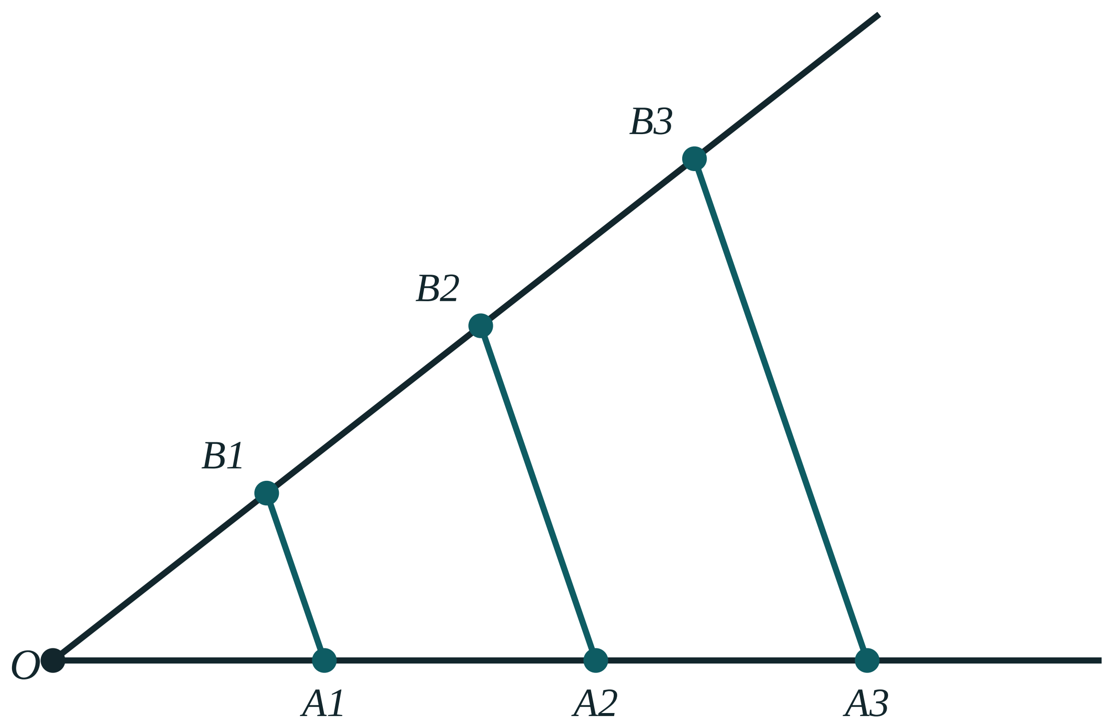
ANSWER

Each part is $\frac{7}{3} \approx 2.33\ cm$ — obtained exactly, without measuring it.

03 Interactive model

Change the angle and the spacing. The ratios in the readout never move.

Thales' theorem



OA_1 **88** | A_1A_2 **88** | OB_1 **88** | B_1B_2 **88** | $OA_1 : OA_2$ **1 : 2** | $OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Thales’ theorem	Fales teoremasi	Теорема Фалеса
Transversal	Kesuvchi	Секущая
Equal segments	Teng kesmalar	Равные отрезки
Proportional	Proporsional	Пропорциональный
Ratio	Nisbat	Отношение
Divide a segment	Kesmani bo‘lish	Разделить отрезок
Construction	Yasash	Построение
Ray	Nur	Луч

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Thales' theorem requires the cutting lines to be:

- equal in length
- parallel
- perpendicular to a transversal
- equally spaced by eye

If parallels cut 5 cm, 5 cm, 5 cm on one transversal and 3 cm on the first piece of another, the second piece is:

- *3 cm*
- *5 cm*
- *9 cm*
- not determined

$OA_1 : OA_2 = 2 : 5$ and $OB_1 = 6$. Then OB_2 is:

- 10
- 15
- 12
- 30

Thales' theorem is most useful for:

- finding areas
- dividing a segment into n equal parts
- proving triangles congruent
- measuring angles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Parallels cut 3 cm, 3 cm on one transversal and 4 cm on the first piece of another. Find the second piece.*

   ANSWER *4 cm*

2. $OA_1 = A_1A_2 = 5$ cm and $OB_1 = 7$ cm. Find OB_2.

   ANSWER *14 cm*

3. *Three parallels cut equal pieces of 6 cm on one line. Find the total for four pieces.*

   ANSWER *24 cm*

4. *Does Thales' theorem apply to three lines that are not parallel?*

   ANSWER No.

5. $OA_1 : OA_2 = 1 : 3$ and $OB_1 = 4$. Find OB_2.

   ANSWER *12*

6. *Equal cuts of 2 cm are made on one transversal. What are the cuts on another?*

   ANSWER Equal to one another, though not necessarily 2 cm.

7. *How many parallels are needed to cut a segment into 4 equal parts?*

   ANSWER *3* (plus the two ends)

Medium · the standard question

7 PROBLEMS

1. $OA_1 = A_1A_2 = A_2A_3 = 4$ cm and $OB_1 = 3$ cm. Find OB_3.

   ANSWER 9 cm

2. $OA_1 = 6$, $OA_2 = 15$, $OB_1 = 8$. Find OB_2.

   ANSWER 20

3. $OA_1 = 4$, $A_1A_2 = 6$, $OB_1 = 10$. Find B_1B_2.

   ANSWER 15

4. A segment of 12 cm is divided into 4 equal parts by Thales' construction. Find each part.

   ANSWER 3 cm

5. $OA_1 : A_1A_2 = 3 : 4$ and $B_1B_2 = 12$. Find OB_1.

   ANSWER 9

6. Describe how to divide a 7 cm segment into 3 equal parts without measuring.

   ANSWER Ray, three equal compass steps, join the last to B, draw parallels.

7. Parallels cut $OA_1 = 5$, $A_1A_2 = 5$, $A_2A_3 = 5$ and $OB_3 = 21$. Find OB_1.

   ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. *Prove Thales' theorem for two equal segments using a parallelogram and ASA.*

ANSWER Through B_1 draw a parallel to the first ray; the parallelogram gives $A_1A_2 = B_1P$, then ASA gives $OB_1 = B_1B_2$.

2. *Divide a segment in the ratio 2 : 3 using Thales' construction.*

ANSWER Step off 5 equal parts on a ray, join the 5th to B, and draw the parallel through the 2nd point.

3. $OA_1 = x$, $A_1A_2 = x + 2$, $OB_1 = 6$, $B_1B_2 = 9$. *Find* x.

ANSWER $\frac{x}{x+2} = \frac{6}{9} \Longrightarrow 9x = 6x + 12 \Longrightarrow x = 4$

4. *In* $\triangle ABC$ *a line parallel to* BC *cuts* AB *at* M *and* AC *at* N, *with* $AM : MB = 2 : 3$ *and* $AN = 8$. *Find* NC.

ANSWER $\frac{AN}{NC} = \frac{2}{3} \Longrightarrow NC = 12$

5. *A ladder rests against a wall and three equally spaced horizontal rungs meet it. Explain why they cut the wall into equal parts.*

ANSWER The rungs are parallel and cut equal pieces on the ladder, so by Thales' theorem they cut equal pieces on the wall.

6. *Show that the construction for n equal parts needs no measurement of AB itself.*

ANSWER Only equal compass steps on the auxiliary ray are used; Thales' theorem transfers the equality to AB.

7. *Three parallels cut a transversal into pieces 3 cm and 5 cm, and cut another transversal into pieces* p *and* $p + 4$. *Find* p.

ANSWER $\frac{3}{5} = \frac{p}{p+4} \Longrightarrow 5p = 3p + 12 \Longrightarrow p = 6$

07 Homework

Homework — 5 problems

Geometry 8, Тема 9, pp. 23–25. Question 5 is a construction — bring it drawn.

1. $OA_1 = A_1A_2 = A_2A_3 = 3$ cm and $OB_1 = 5$ cm. Find OB_3.
2. $OA_1 = 8$, $OA_2 = 20$, $OB_1 = 6$. Find OB_2.
3. $OA_1 : A_1A_2 = 2 : 5$ and $B_1B_2 = 20$. Find OB_1.
4. In $\triangle ABC$, $MN \parallel BC$ with $AM : MB = 3 : 4$ and $AN = 9$. Find NC.
5. *Construct the division of a 9 cm segment into 5 equal parts and state the theorem you used.*

Applications of Thales' theorem

The theorem inside a triangle: proportional segments, the line through a midpoint, and the measurements you cannot reach with a tape.

LESSON 13–14 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 9–10 | STAGE 9 · 13.4

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Apply Thales' theorem inside a triangle to find unknown lengths.
- Prove that a line through the midpoint of one side, parallel to a second, bisects the third.
- Divide a segment in a given ratio by construction.
- Solve practical measuring problems using proportional segments.

TEXTBOOK

National	Темы 9–10, pp. 23–28
Cambridge	Learner's Book pp. 293–299
Workbook	Workbook 13.4

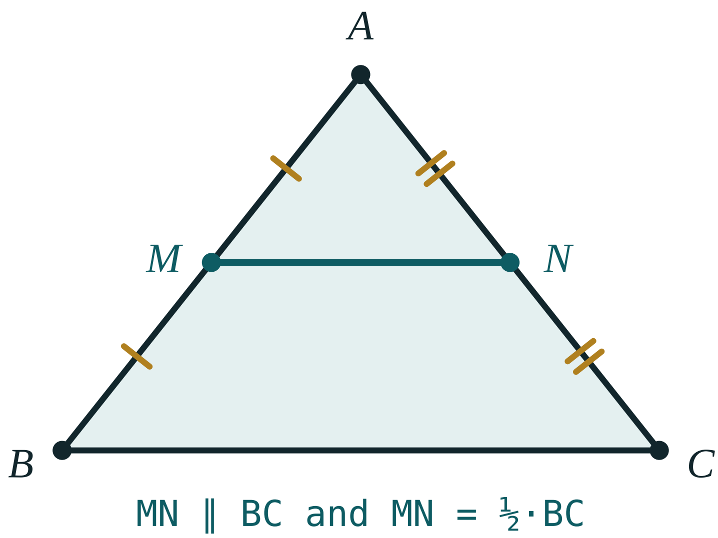
01 Explanation

Thales inside a triangle

THE FORM YOU WILL USE MOST

If a line parallel to BC cuts AB at M and AC at N, then

$$\frac{AM}{MB} = \frac{AN}{NC} \quad \text{and} \quad \frac{AM}{AB} = \frac{AN}{AC}$$



When M and N are the midpoints, the ratio is 1 : 1 — the special case that gives the midline.

Two different ratios, and mixing them up is the commonest error in this topic. Decide first whether you are comparing a **part with a part** ($AM : MB$) or a **part with the whole** ($AM : AB$), and stay with that choice on both sides.

The midpoint corollary

COROLLARY

A line drawn through the midpoint of one side of a triangle, parallel to a second side, bisects the third side.

Proof. With $AM = MB$ the ratio $\frac{AM}{MB}$ is 1, so $\frac{AN}{NC}$ is also 1 and $AN = NC$. ∎

This corollary is what makes the midline theorem of the next lesson possible, so it is worth stating separately.

Dividing a segment in a given ratio

To divide AB in the ratio $2 : 3$:

1. Draw a ray from A and step off $2 + 3 = 5$ equal segments.
2. Join the 5th point to B.
3. Through the 2nd point draw a parallel to that line.
4. It meets AB at the point dividing it in the ratio $2 : 3$.

The same construction, with n equal steps and a parallel through each, divides the segment into n equal parts.

Measuring what you cannot reach

The practical use of the theorem is measuring across an obstacle — a river, a building, a fenced field. Set up two similar configurations sharing an angle, measure the three lengths you can reach, and solve the proportion for the fourth.

WORKED PATTERN

A pole $1.5\ m$ tall casts a shadow $2\ m$ long; at the same moment a tree casts a shadow $14\ m$ long. The sun's rays are parallel, so

$$\frac{1.5}{2} = \frac{h}{14} \implies h = 10.5\ m$$

02 Worked examples

EXAMPLE 1 ***In △ABC, MN ∥ BC with AM = 4, MB = 6 and AN = 6. Find NC.***

1. $$\frac{AM}{MB} = \frac{AN}{NC}$$

   Part with part on both sides.

2. $$\frac{4}{6} = \frac{6}{NC}$$

   Substitute.

3. $4 \cdot NC = 36$

   Cross-multiply.

4. $NC = 9$

ANSWER

$NC = 9$

EXAMPLE 2 ***In $\triangle ABC$, $MN \parallel BC$ with $AM = 3$, $AB = 12$ and $AC = 20$. Find AN.***

1. $$\frac{AM}{AB} = \frac{AN}{AC}$$
   This time it is part with whole — note AB, not MB.

2. $$\frac{3}{12} = \frac{AN}{20}$$

3. $12 \cdot AN = 60$

4. $AN = 5$

ANSWER

$AN = 5$

EXAMPLE 3 **A 1.8 m man casts a 2.4 m shadow. A tower casts a 40 m shadow at the same time. Find its height.**

1. The sun's rays are parallel, so the two right triangles have proportional sides.
   That is Thales in disguise.

2. $$\frac{1.8}{2.4} = \frac{h}{40}$$
   Height over shadow, on both sides.

3. $2.4h = 72$

4. $h = 30\ m$

ANSWER

$30\ m$

03 Interactive model

Move the spacing slider so the cuts are unequal, and check that the ratios still match on both rays.

Which proportion do I write?

> Given as **part and part** on the first side.

1. The data give

AM and

MB — two pieces of the same side.

So use part-to-part.

2. $$\frac{AM}{MB} = \frac{AN}{NC} \Longrightarrow \frac{4}{6} = \frac{6}{NC}$$

3. $4 \cdot NC = 36, \text{ so } NC = 9$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Proportion	Proporsiya	Пропорция
Cross-multiply	Krest-nakrest ko‘paytirish	Умножить крест-накрест
Midpoint	O‘rta nuqta	Середина
Scale	Masshtab	Масштаб
Shadow	Soya	Тень
Similar triangles	O‘xshash uchburchaklar	Подобные треугольники
Part to whole	Qism-butunga nisbat	Отношение части к целому
Part to part	Qism-qismga nisbat	Отношение части к части

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$MN \parallel BC$ with $AM = 2$, $MB = 3$, $AN = 4$. Then NC is:

- 6
- 5
- 8
- 2.4

$MN \parallel BC$ with $AM = 2$, $AB = 5$, $AC = 15$. Then AN is:

- 6
- 7.5
- 10
- 3

A line through the midpoint of AB parallel to BC meets AC at N. Then N is:

- a third of the way along AC
- the midpoint of AC
- not determined
- the same point as C

A 2 m pole casts a 3 m shadow. A 24 m shadow belongs to an object of height:

- *16 m*
- *36 m*
- *12 m*
- *48 m*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $MN \parallel BC$, $AM = 2$, $MB = 3$, $AN = 4$. Find NC.

   ANSWER 6

2. $MN \parallel BC$, $AM = 5$, $MB = 5$, $AN = 7$. Find NC.

   ANSWER 7

3. $MN \parallel BC$, $AM = 3$, $AB = 9$, $AC = 12$. Find AN.

   ANSWER 4

4. *A 2 m pole casts a 3 m shadow; a tree casts 24 m. Find its height.*

   ANSWER 16 m

5. M is the midpoint of AB and $MN \parallel BC$. What is N?

   ANSWER The midpoint of AC.

6. $MN \parallel BC$ and $AM : MB = 1 : 2$. Find $AN : NC$.

   ANSWER $1 : 2$

7. $MN \parallel BC$, $AM = 4$, $AN = 4$, $NC = 8$. Find MB.

   ANSWER 8

Medium · the standard question

7 PROBLEMS

1. $MN \parallel BC$, $AM = 4$, $MB = 6$, $AN = 6$. Find NC.

   ANSWER 9

2. $MN \parallel BC$, $AM = 3$, $AB = 12$, $AC = 20$. Find AN.

   ANSWER 5

3. *A 1.8 m man casts a 2.4 m shadow; a tower casts 40 m. Find its height.*

   ANSWER 30 m

4. $MN \parallel BC$, $AN : NC = 3 : 5$ and $AB = 24$. Find AM.

   ANSWER 9

5. *Describe how to divide a segment in the ratio 2 : 3.*

   ANSWER 5 equal steps on a ray, join the 5th to B, draw the parallel through the 2nd.

6. $MN \parallel BC$, $AM = x$, $MB = 8$, $AN = 6$, $NC = 12$. Find x.

   ANSWER $x = 4$

7. $MN \parallel BC$ and MN cuts off a triangle with $AM : AB = 2 : 7$. Find $AN : AC$.

   ANSWER $2 : 7$

Hard · stretch and reason

7 PROBLEMS

1. *Prove that a line through the midpoint of AB parallel to BC bisects AC.*

   ANSWER $\frac{AM}{MB} = 1$ forces $\frac{AN}{NC} = 1$ by Thales, so $AN = NC$.

2. *In $\triangle ABC$, $MN \parallel BC$, $AM = x$, $MB = x + 3$, $AN = 4$, $NC = 8$. Find x.*

   ANSWER $\frac{x}{x+3} = \frac{4}{8} \Longrightarrow 2x = x + 3 \Longrightarrow x = 3$

3. *A river is crossed by sighting: two parallel lines cut a base line into 12 m and 18 m, and one side of the river measures 20 m. Find the far distance.*

   ANSWER $\frac{12}{18} = \frac{20}{d} \Longrightarrow d = 30\ m$

4. *$MN \parallel BC$ with $AM : MB = 3 : 4$. The perimeter of $\triangle AMN$ is 21. Find the perimeter of $\triangle ABC$.*

   ANSWER Sides scale by $\frac{3}{7}$, so the perimeter of $\triangle ABC$ is 49.

5. *Divide a segment into 7 equal parts and explain why no measurement of its length is needed.*

   ANSWER Seven equal compass steps on a ray plus parallels transfer the equality by Thales; the length of AB is never used.

6. *In trapezium ABCD ($AD \parallel BC$), the diagonals meet at O. Prove* $\frac{AO}{OC} = \frac{AD}{BC}$.

   ANSWER The parallels AD and BC cut the transversals AC and BD proportionally.

7. *Three parallels cut one transversal into 4 cm and 6 cm and another into pieces whose sum is 20 cm. Find them.*

   ANSWER Ratio $2 : 3$ of 20 gives $8\ cm$ and $12\ cm$.

07 Homework

Homework — 6 problems

Geometry 8, Темы 9–10, pp. 23–28. Write the proportion before you substitute.

1. $MN \parallel BC$, $AM = 6$, $MB = 9$, $AN = 8$. Find NC.
2. $MN \parallel BC$, $AM = 5$, $AB = 15$, $AC = 21$. Find AN.
3. *A 1.6 m person casts a 2 m shadow; a flagpole casts 15 m. Find its height.*
4. $MN \parallel BC$ and $AM : MB = 2 : 5$. Find $AN : AC$.
5. *Divide a 10 cm segment in the ratio 3 : 2 by construction.*
6. In $\triangle ABC$, $MN \parallel BC$ and the perimeter of $\triangle AMN$ is half that of $\triangle ABC$. Find $AM : MB$.

Home / Grade 8 / Geometry / Quarter 1

The midline of a triangle and of a trapezium

Two theorems with one idea: join the midpoints and you get a segment parallel to the base, of exactly the average length.

LESSON 15 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 10–11 | STAGE 9 · 13.2

LESSON CLOCK

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State and prove the midline theorem for a triangle.
- State and use the midline theorem for a trapezium.
- See the triangle case as the trapezium case with $b = 0$.
- Use midlines to solve length problems and to prove other results.

TEXTBOOK

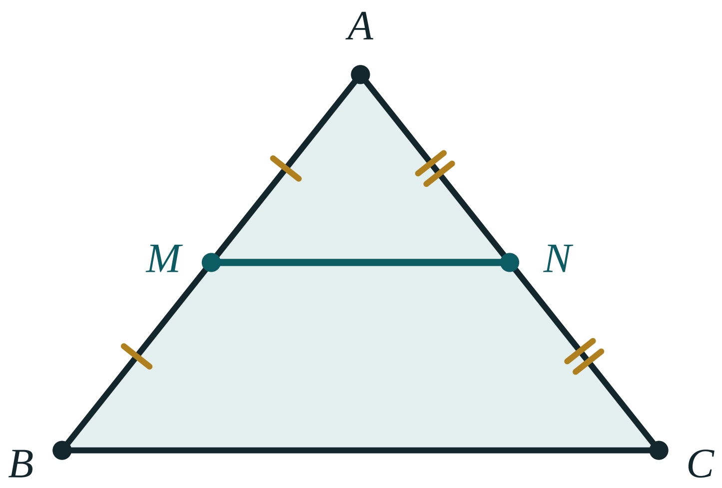
National	Темы 10–11, pp. 26–28
Cambridge	Learner’s Book pp. 281–284
Workbook	Workbook 13.2

01 Explanation

The midline of a triangle

DEFINITION

The **midline** of a triangle is the segment joining the midpoints of two of its sides.



MN ∥ BC and MN = ½·BC

M and N are the midpoints of AB and AC. The midline MN is parallel to BC and exactly half its length.

THEOREM

The midline of a triangle is parallel to the third side and equal to half of it:

$$MN \parallel BC \quad \text{and} \quad MN = \frac{1}{2} BC$$

Proof. Extend MN beyond N to a point P with $NP = MN$. Then $\triangle ANM \cong \triangle CNP$ by SAS ($AN = NC$, $MN = NP$, vertically opposite angles at N). Hence $CP = AM = MB$ and $\angle CPN = \angle AMN$, which are alternate angles — so $CP \parallel MB$. Now $MBCP$ has one pair of sides equal *and* parallel, so by Test 1 it is a parallelogram. Therefore $MP \parallel BC$ and $MP = BC$; and since $MN = \frac{1}{2} MP$, the theorem follows. ∎

CONSEQUENCE

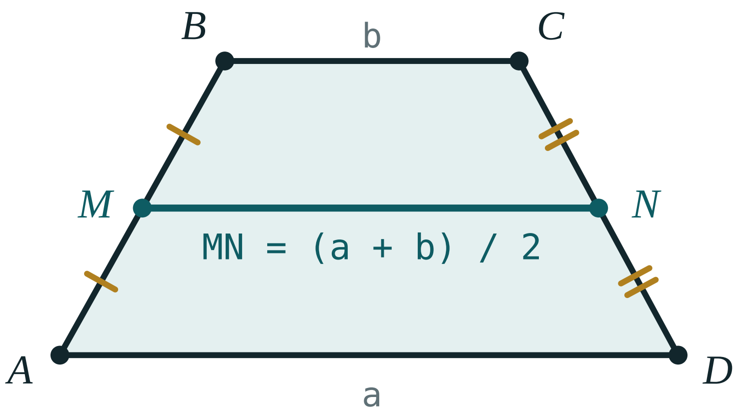
The three midlines of a triangle cut it into four congruent triangles, each similar to the original with half the side lengths and a quarter of the area. The middle triangle has perimeter equal to half the original.

The midline of a trapezium

THEOREM

The midline of a trapezium — the segment joining the midpoints of the two legs — is parallel to the bases and equal to their half-sum:

$$MN \parallel AD \parallel BC \quad \text{and} \quad MN = \frac{a+b}{2}$$



The midline of a trapezium is the average of the two bases.

Proof. Draw the diagonal BD and let it meet MN at K. In $\triangle ABD$, MK joins the midpoints of AB and BD — a midline, so $MK = \frac{1}{2} AD = \frac{a}{2}$. In $\triangle BCD$, KN is a midline, so $KN = \frac{1}{2} BC = \frac{b}{2}$. Adding gives $MN = \frac{a+b}{2}$. ∎

ONE THEOREM, NOT TWO

Let the shorter base shrink to a point, $b = 0$. The trapezium becomes a triangle and the formula becomes $MN = \frac{a}{2}$ — the triangle midline theorem. The triangle case is the trapezium case with one base of length zero.

Using midlines

Two habits that solve most midline problems:

- When you see **two midpoints**, join them — you have created a midline.
- When you see **one midpoint** and want a second, draw a parallel to a side; the midpoint corollary from the last lesson supplies it.

Area is also useful: the area of a trapezium is $S = MN \cdot h$ — the midline multiplied by the height.

02 Worked examples

EXAMPLE 1 ***In △ABC, BC = 18 cm. M and N are the midpoints of AB and AC. Find MN.***

1. MN joins two midpoints — it is a midline.

2. $MN = \frac{1}{2} BC$

   Midline theorem.

3. $MN = 9\ cm$

ANSWER

$MN = 9\ cm$

EXAMPLE 2 **A trapezium has bases 8 cm and 14 cm and height 5 cm. Find its midline and area.**

1. $MN = \frac{8 + 14}{2} = 11\ cm$

   Half-sum of the bases.

2. $S = MN \cdot h = 11 \cdot 5$

   Area of a trapezium.

3. $S = 55\ cm^2$

ANSWER

$MN = 11\ cm, S = 55\ cm^2$

EXAMPLE 3

The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both bases.

1. $a + b = 2 \cdot 12 = 24$

   From the midline formula.

2. $a = b + 3$

   The given relation.

3. $b + 3 + b = 24 \Longrightarrow 2b = 21$

4. $b = 10.5,\ a = 13.5$

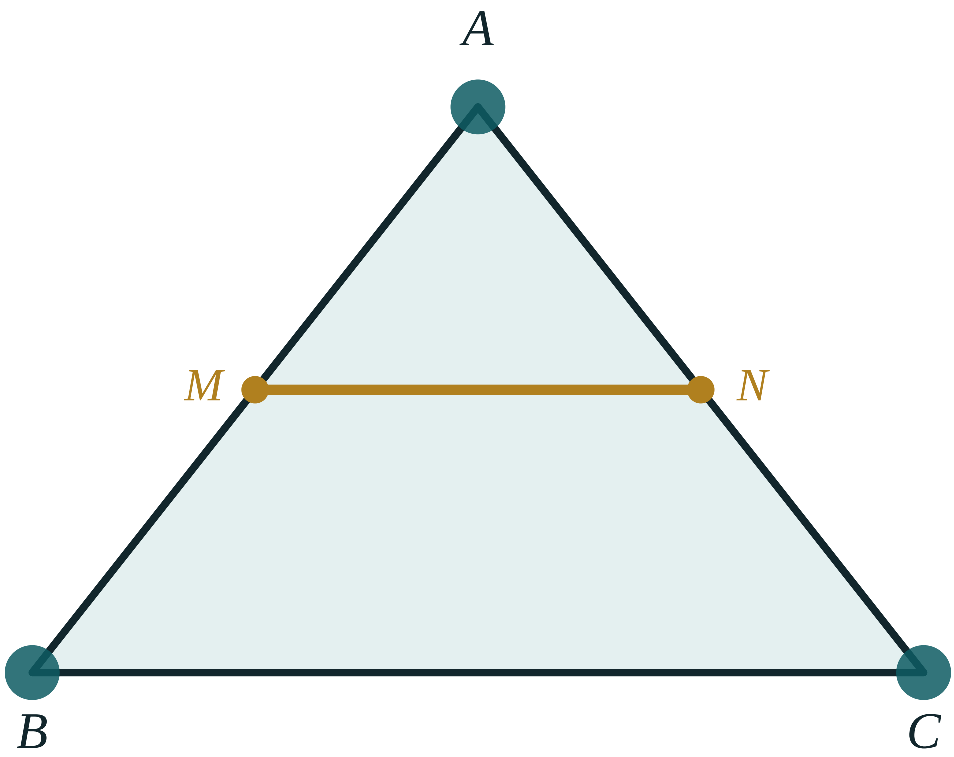
ANSWER

$10.5\ cm$ and $13.5\ cm$

03 Interactive model

Drag a vertex to a wild position — the ratio BC : MN stays at 2.00 throughout.

The midline of a triangle



BC **260**	MN **130**	BC / MN **2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Midline of a triangle	Uchburchakning o‘rta chizig‘i	Средняя линия треугольника
Midline of a trapezium	Trapetsiyaning o‘rta chizig‘i	Средняя линия трапеции
Midpoint	O‘rta nuqta	Середина
Half-sum	Yarim yig‘indi	Полусумма
Corollary	Natija	Следствие
Congruent triangles	Teng uchburchaklar	Равные треугольники
Parallel	Parallel	Параллельный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The midline of a triangle with base 20 cm is:

- *40 cm*
- *10 cm*
- *20 cm*
- *5 cm*

The midline of a trapezium with bases 7 and 13 is:

- *20*
- *10*
- *6*
- *91*

The three midlines of a triangle cut it into:

- 2 triangles
- 3 triangles
- 4 congruent triangles
- 6 triangles

A trapezium has midline 9 and height 6. Its area is:

- *54*
- *27*
- *15*
- *108*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *In $\triangle ABC$, $BC = 18$ cm. Find the midline MN parallel to BC.*

   ANSWER *9 cm*

2. *A midline of a triangle is 7 cm. Find the side it is parallel to.*

   ANSWER *14 cm*

3. *A trapezium has bases 6 cm and 10 cm. Find the midline.*

   ANSWER *8 cm*

4. *A trapezium has bases 5 cm and 15 cm. Find the midline.*

   ANSWER *10 cm*

5. *A trapezium has midline 9 cm and height 6 cm. Find the area.*

   ANSWER *54 cm^2*

6. *A triangle has sides 10, 12, 14. Find the perimeter of the triangle formed by its midlines.*

   ANSWER *18*

7. *The midline of a trapezium is 11 cm and one base is 8 cm. Find the other.*

   ANSWER *14 cm*

Medium · the standard question

7 PROBLEMS

1. *A trapezium has bases 8 cm and 14 cm and height 5 cm. Find the midline and area.*

   ANSWER *11 cm, 55 cm²*

2. *The midline of a trapezium is 12 cm and one base is 3 cm shorter than the other. Find both.*

   ANSWER *10.5 cm* and *13.5 cm*

3. *In $\triangle ABC$ the midlines have lengths 4, 5, 6. Find the sides of $\triangle ABC$.*

   ANSWER *8, 10, 12*

4. *A trapezium has area 60 cm² and height 5 cm. Find its midline.*

   ANSWER *12 cm*

5. *The midline of a trapezium is 10 cm and the bases are in the ratio 2 : 3. Find them.*

   ANSWER *8 cm* and *12 cm*

6. *In $\triangle ABC$, M and N are the midpoints of AB and AC, and $MN = 6.5$ cm. Find BC.*

   ANSWER *13 cm*

7. *A trapezium has bases 9 and 21 and legs 10 and 10. Find the midline and the height.*

   ANSWER Midline *15*; $\frac{21-9}{2} = 6$, so $h = 8$

Hard · stretch and reason

7 PROBLEMS

1. *Prove the midline theorem for a triangle.*

   ANSWER Extend MN to P with $NP = MN$; $\triangle ANM \cong \triangle CNP$ by SAS makes $MBCP$ a parallelogram, so $MP = BC$ and MN is half of it.

2. *Prove that the midline of a trapezium equals the half-sum of the bases.*

   ANSWER The diagonal BD splits MN into two triangle midlines of lengths $\frac{a}{2}$ and $\frac{b}{2}$.

3. *The midlines of a triangle cut it into four triangles. Compare their areas with the original.*

   ANSWER Each is $\frac{1}{4}$ of the original area, and all four are congruent.

4. *The midline of a trapezium is 14 cm and the bases are in the ratio 3 : 4. Find both bases.*

   ANSWER $a + b = 28$ in the ratio $3 : 4$: 12 cm and 16 cm.

5. *P, Q, R, S are the midpoints of the sides of any quadrilateral. Prove PQRS is a parallelogram.*

   ANSWER PQ and SR are both midlines parallel to the diagonal AC and equal to half of it.

6. *The midline of a triangle divides it into a triangle and a trapezium. Compare their areas.*

   ANSWER The small triangle is $\frac{1}{4}$ of the whole, so the trapezium is $\frac{3}{4}$ — a ratio of $1 : 3$.

7. *A trapezium has midline 8 and the two bases differ by 4. Find the area if the height is 7.*

   ANSWER Bases 6 and 10; $S = 8 \cdot 7 = 56$.

07 Homework

Homework — 5 problems

Geometry 8, Темы 10–11, pp. 26–28.

1. *In $\triangle ABC$, $BC = 22$ cm. Find the midline parallel to BC.*
2. *A trapezium has bases 11 cm and 17 cm. Find the midline, then the area if the height is 6 cm.*

3. *The midline of a trapezium is 15 cm and one base is 4 cm longer than the other. Find both.*
4. *The midlines of a triangle are 6, 7 and 8. Find the perimeter of the triangle.*
5. *Prove that the midline of a triangle is parallel to the third side.*

Practical tasks — ruler-and-compass constructions

The national “practical tasks” lesson, carrying the Cambridge Stage 9 constructions the national programme does not cover.

LESSON 16 | 1 × 40 MIN | GEOMETRY 8, TEMA 12 | STAGE 9 · 5.4 [CAMBRIDGE INSERT]

LESSON CLOCK

4' | 10' | 10' | 14' | 2'

Warm-up	4 min
Explanation	10 min
Construct	10 min
Practice	14 min
Homework	2 min

LEARNING OBJECTIVES

- Construct the perpendicular bisector of a segment and explain why it works.
- Construct the bisector of an angle.
- Construct a perpendicular from a point to a line, and an angle of 60° and 30°.
- Use a construction to solve a practical placement problem.

TEXTBOOK

National	Тема 12, pp. 29–32
Cambridge	Learner’s Book pp. 119–123
Workbook	Workbook 5.4

01 Explanation

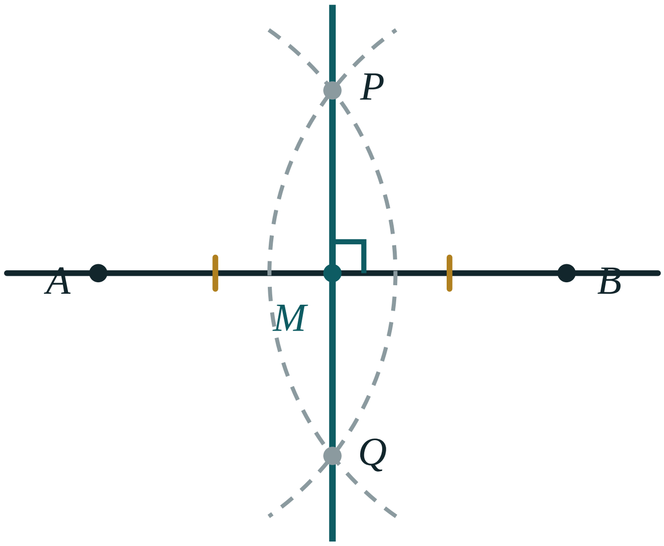
The rules of the game

ALLOWED

A straight edge, to draw a line through two points, and compasses, to draw a circle of a chosen radius about a chosen point. **Not allowed:** measuring lengths with the ruler’s scale, or angles with a protractor.

Leave every construction arc on the page. The arcs are the working, and in Cambridge papers they carry the marks — a correct figure with the arcs rubbed out scores less than a slightly wobbly one with them showing.

Construction 1 — the perpendicular bisector of AB

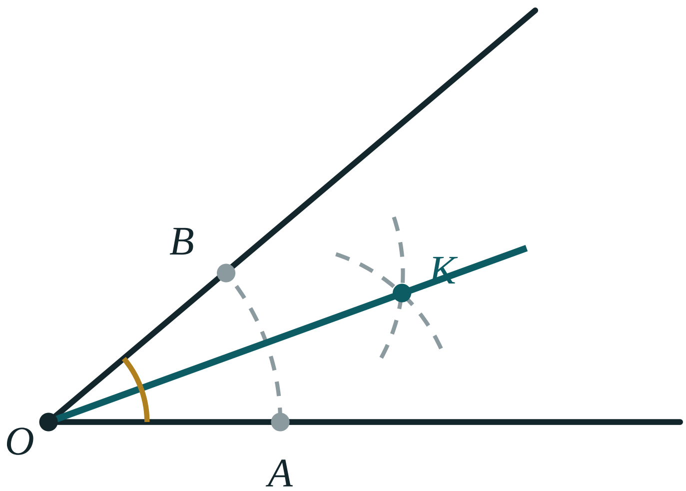


Equal radii from A and from B meet at P and Q. The line PQ is the perpendicular bisector of AB.

1. Open the compasses to more than half of AB.
2. Draw an arc centred at A, and another of the **same radius** centred at B.
3. They meet at two points, P and Q. Join them.

Why it works. P and Q are each the same distance from A and from B, so both lie on the set of points equidistant from A and B — which is exactly the perpendicular bisector. The line also gives the **midpoint** M for free.

Construction 2 — the bisector of an angle



One arc from O cuts the arms at A and B; equal arcs from A and B meet at K; OK is the bisector.

1. With centre O draw an arc cutting the arms at A and B.

2. With the same radius, draw arcs centred at A and at B; call their intersection K.
3. Draw OK.

Why it works. $OA = OB$ and $AK = BK$ and OK is common, so $\triangle OAK \cong \triangle OBK$ by SSS, and therefore $\angle AOK = \angle KOB$. ∎

Two more you should be able to do

A perpendicular from a point P to a line ℓ. Draw an arc centred at P cutting ℓ at two points, then construct the perpendicular bisector of the segment between them — it passes through P.

An angle of 60°. Draw an arc centred at O cutting the line at A; with the same radius, an arc centred at A cuts the first at B. Then $\triangle OAB$ is equilateral, so $\angle AOB = 60°$. Bisect it for $30°$, bisect again for $15°$.

WHERE THESE ARE USED

Every one of these is a **locus**: the perpendicular bisector is the set of points equidistant from two points; the angle bisector is the set of points equidistant from two lines. That is why they answer questions like “where should the well be dug so that it is the same distance from both villages?”

02 Worked examples

EXAMPLE 1 Find the point equidistant from three villages A, B and C.

1. Construct the perpendicular bisector of AB.
   Every point on it is equidistant from A and B.
2. Construct the perpendicular bisector of BC.
   Every point on it is equidistant from B and C.
3. They meet at one point O.
   $OA = OB$ and $OB = OC$, so all three are equal.
4. O is the centre of the circle through A, B, C.
   The circumcentre of the triangle.

ANSWER

The intersection of the perpendicular bisectors — the circumcentre.

EXAMPLE 2 Construct an angle of 45° using compasses only.

1. Construct a perpendicular to a line at a point O.
   That gives $90°$.
2. Bisect the right angle.
   Two equal halves of $90°$.
3. Each half is $45°$.
   No protractor was used at any stage.

ANSWER

Perpendicular, then bisect.

EXAMPLE 3 **A path must be the same distance from two straight roads that cross. Where does it run?**

1. The set of points equidistant from two lines is the bisector of the angle between them.
2. Two crossing roads make two pairs of vertically opposite angles.
   So there are two bisectors.
3. The two bisectors are perpendicular to each other.
   Both answers are valid paths.

ANSWER

Along either bisector of the angles between the roads.

03 Interactive model

Do these on the board with real compasses first, then use the step-through to check the reasoning.

Construction steps

Given a segment AB.

1. Open the compasses to more than half of AB.

   Any radius under half gives no intersection.

2. Arc centred at A; arc of the **same** radius centred at B.

   The equal radius is the whole point.

3. They meet at P and Q. Draw the line PQ.

4 $PA = PB$

and $QA = QB$

, so PQ

is A
the
locus
of
points
equidistant
from

and B

. It is perpendicular to AB and passes through its midpoint.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Compasses	Sirkul	Циркуль
Straight edge	Chizg‘ich	Линейка
Arc	Yoy	Дуга
Radius	Radius	Радиус
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Angle bisector	Burchak bissektrisasi	Биссектриса угла
Locus	Geometrik o‘rin	Геометрическое место точек
Equidistant	Teng uzoqlikda	Равноудалённый
Circumcentre	Tashqi aylana markazi	Центр описанной окружности
Incentre	Ichki aylana markazi	Центр вписанной окружности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For the perpendicular bisector construction the compass radius must be:

- exactly half of AB
- more than half of AB
- less than half of AB
- any radius at all

The set of points equidistant from two points is:

- a circle
- the perpendicular bisector of the segment joining them
- an angle bisector
- a parabola

The set of points equidistant from two crossing lines is:

- a circle
- the perpendicular bisector
- the pair of angle bisectors
- a single point

To construct 30° you first construct:

- *45°*
- *60°* and bisect it
- *90°* and trisect it
- *15°* and double it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Construct the perpendicular bisector of a 6 cm segment.*

   ANSWER Equal arcs from both ends; join the two intersections.

2. *Construct the midpoint of a 7 cm segment.*

   ANSWER It is where the perpendicular bisector crosses.

3. *Construct the bisector of a $70°$ angle. What is each half?*

   ANSWER $35°$

4. *Construct an angle of $60°$.*

   ANSWER Equilateral triangle by equal compass widths.

5. *Construct an angle of $30°$.*

   ANSWER Construct $60°$ and bisect it.

6. *What is the locus of points equidistant from A and B?*

   ANSWER The perpendicular bisector of AB.

7. *What is the locus of points 3 cm from a point O?*

   ANSWER A circle of radius 3 cm centred at O.

Medium · the standard question

7 PROBLEMS

1. *Construct a perpendicular from a point P to a line ℓ.*

   ANSWER Arc from P cutting ℓ twice, then the perpendicular bisector of that segment.

2. *Construct an angle of 45°.*

   ANSWER $90°$ then bisect.

3. *Construct an angle of 15°.*

   ANSWER $60° \to 30° \to 15°$ by bisecting twice.

4. *Construct the circumcentre of a triangle.*

   ANSWER Intersection of two perpendicular bisectors.

5. *Construct the incentre of a triangle.*

   ANSWER Intersection of two angle bisectors.

6. *Explain why the angle-bisector construction works.*

   ANSWER $\triangle OAK \cong \triangle OBK$ by SSS.

7. *Construct a triangle with sides 5 cm, 6 cm and 7 cm.*

   ANSWER Draw the 7 cm base, then arcs of 5 cm and 6 cm from its ends.

Hard · stretch and reason

7 PROBLEMS

1. *A well must be equidistant from three houses. Where is it, and when is there no such point?*

   ANSWER At the circumcentre; there is none if the three houses lie in a straight line.

2. *Construct an angle of 75°.*

   ANSWER $60° + 15°$ — build $60°$, then bisect $30°$ onto it.

3. *A treasure is 4 cm from point A and equidistant from B and C. Describe the construction.*

   ANSWER Circle of radius 4 cm about A, intersected with the perpendicular bisector of BC — up to two positions.

4. *Prove that the perpendicular bisectors of the three sides of a triangle are concurrent.*

   ANSWER The intersection of two of them is equidistant from all three vertices, so it lies on the third as well.

5. *Construct the tangent-length point: divide a 9 cm segment into 3 equal parts by construction.*

   ANSWER Thales' construction: three equal steps on a ray, join the last to B, draw parallels.

6. *Construct an angle of 105° and explain your steps.*

   ANSWER $90° + 15°$: build a perpendicular, then bisect $60°$ twice to get $15°$ and add it.

7. *A path must be equidistant from two roads and 5 cm from a crossroads. Find the possible positions.*

   ANSWER Intersect the two angle bisectors with the circle of radius 5 cm — up to four points.

07 Homework

Homework — 5 constructions

Geometry 8, Тема 12, pp. 29–32, and Cambridge Learner's Book 5.4. Leave all arcs visible.

1. *Construct the perpendicular bisector of an 8 cm segment.*
2. *Construct the bisector of an angle of about 100° and measure each half to check.*
3. *Construct angles of 60°, 30° and 15° from the same line.*
4. *Construct a triangle with sides 4 cm, 6 cm and 7 cm, then its circumcentre.*

5. A house is to be built equidistant from two straight roads and 6 cm from a marked tree. Construct all possible positions.

Home / Grade 8 / Geometry / Quarter 1

Chapter revision — angles in polygons

The whole of Chapter I pulled together, in the shape of Cambridge Project 2, “Angle tangle”.

LESSON 17 | 1 × 40 MIN | GEOMETRY 8, ТЕСТ 1 | STAGE 9 · PROJECT 2 [CAMBRIDGE INSERT]

LESSON CLOCK

4' | 10' | 8' | 16' | 2'

Recall round	4 min
Project 2	10 min
Interactive	8 min
Mixed practice	16 min
Homework	2 min

LEARNING OBJECTIVES

- Recall the definitions, properties and tests of Chapter I without notes.
- Choose the right property for a given problem.
- Work the Cambridge “Angle tangle” investigation.
- Identify personal gaps before the control work.

TEXTBOOK

National	Тест 1, p. 33
Cambridge	Learner’s Book p. 127
Workbook	Workbook 5.2–5.3

01 Explanation

The recall round — four minutes, no notes

Ask these in order, round the class. Anything that stalls is what tomorrow’s control work will catch.

1. Sum of the interior angles of an n-gon? $180°(n-2)$
2. Sum of the exterior angles? $360°$, always.
3. Three properties of a parallelogram? Opposite sides equal, opposite angles equal, diagonals bisect each other.
4. Three tests for a parallelogram? One pair equal and parallel; both pairs of sides equal; diagonals bisect each other.
5. What is special about the diagonals of a rectangle? They are equal.
6. Of a rhombus? Perpendicular, and they bisect the angles.
7. Of a square? All of the above.
8. Midline of a triangle? $MN \parallel BC$, $MN = \frac{1}{2} BC$.
9. Midline of a trapezium? $\frac{a+b}{2}$.

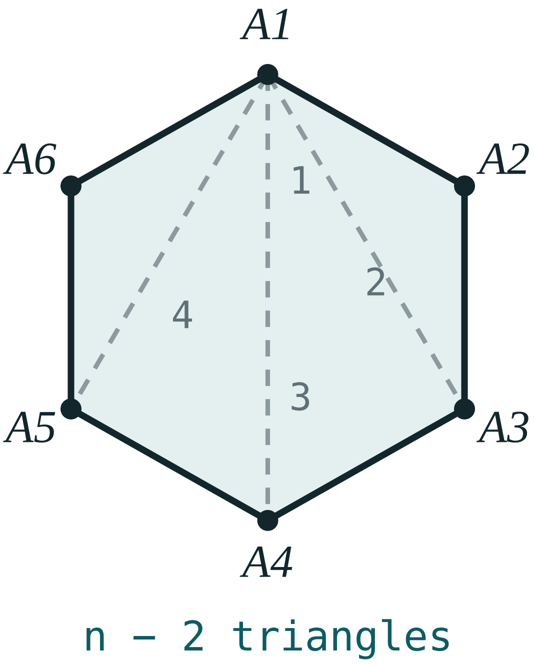
.0. Thales' theorem? Parallel lines cut proportional segments on any transversals.

Project 2 · Angle tangle

The Cambridge investigation, adapted to the national chapter. Work in pairs.

THE TASK

1. Draw any convex pentagon and measure its five interior angles. Add them. Compare with $180° \cdot 3 = 540°$.
2. Now draw a **non-convex** pentagon and repeat. Does the formula still hold?
3. Extend each side in turn and measure the exterior angles. What do they add to?
4. Predict the answers for a 9-gon *before* drawing it. Then check.
5. Write one sentence explaining why the exterior sum does not depend on n.



The diagonals from one vertex are what turn the polygon into triangles — the heart of the whole investigation.

Expect the surprise at step 2: the interior formula survives for non-convex polygons (with reflex angles counted properly), but the exterior-angle argument does not, because "turning through the exterior angle" goes backwards at a reflex vertex.

The three traps to name out loud

BEFORE THE CONTROL WORK

1. **"The diagonals of a parallelogram are equal."** They bisect each other. Equal needs a rectangle.
2. **"Equal diagonals mean a rectangle."** Only if the shape is already a parallelogram — an isosceles trapezium has equal diagonals too.
3. **Mixing part-to-part with part-to-whole** in a Thales proportion. Decide which one and stay with it on both sides.

02 Worked examples

EXAMPLE 1 **A regular polygon has interior angle 150°. Find n, and the sum of its interior angles.**

1. $exterior = 180° - 150° = 30°$

   Work with the exterior angle.

2. $n = \dfrac{360°}{30°} = 12$

3. $sum = 180°(12 - 2) = 1800°$

   Check with the interior formula.

ANSWER

$n = 12$, sum $1800°$

EXAMPLE 2 ***ABCD is a parallelogram with AC = BD. What kind of quadrilateral is it, and why?***

1. It is already a parallelogram, so the diagonals bisect each other.
2. Equal diagonals plus bisecting gives four equal half-diagonals.
3. $\triangle ABC \cong \triangle BAD$ by SSS, so $\angle A = \angle B$.
4. They are supplementary, so each is $90°$ — a rectangle.

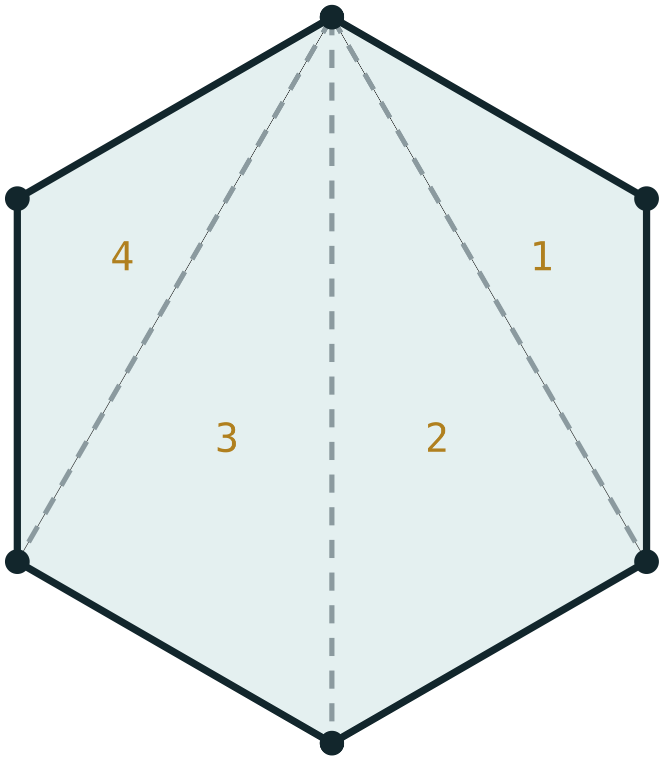
ANSWER

A rectangle.

03 Interactive model

Run the polygon model up to 12 sides while the class checks their Project 2 predictions.

Angle sum of a polygon



$$180° \times (6 - 2) = 720°$$

sides n **6** | triangles **4** | interior sum **720°** | each angle **120°** | exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Investigation	Tadqiqot	Исследование
Predict	Bashorat qilish	Предсказать
Reflex angle	Yoyiq burchakdan katta burchak	Рефлексный угол
Justify	Asoslash	Обосновать
Regular polygon	Muntazam ko‘pburchak	Правильный многоугольник
Family tree	Tasnif sxemasi	Схема классификации

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A regular polygon has interior angle 150°. It has:

- 10 sides
- 12 sides
- 15 sides
- 20 sides

A parallelogram with equal diagonals is:

- a rhombus
- a rectangle
- a trapezium
- a kite

A parallelogram with perpendicular diagonals is:

- a rhombus
- a rectangle
- a trapezium
- not determined

The midline of a trapezium with bases 9 and 15 is:

- *6*
- *12*
- *24*
- *3*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Sum of the interior angles of a nonagon (9 sides).*

   ANSWER *1260°*

2. *Each exterior angle of a regular octagon.*

   ANSWER *45°*

3. *In parallelogram ABCD, ∠A = 55°. Find ∠B.*

   ANSWER *125°*

4. *The diagonals of a rhombus are 8 and 6. Find the area.*

   ANSWER *24*

5. *A rectangle has sides 5 and 12. Find the diagonal.*

   ANSWER *13*

6. *A trapezium has bases 9 and 15. Find the midline.*

   ANSWER *12*

7. *In △ABC, BC = 16. Find the midline parallel to BC.*

   ANSWER *8*

Medium · the standard question

7 PROBLEMS

1. *A regular polygon has interior angle 150°. Find n and the interior sum.*

   ANSWER $n = 12$, $1800°$

2. *ABCD is a parallelogram with $AC = BD$. Name the shape and justify.*

   ANSWER A rectangle — equal and bisecting diagonals force right angles.

3. *A rhombus has side 10 and one diagonal 12. Find the other diagonal and the area.*

   ANSWER 16, area 96

4. *An isosceles trapezium has bases 6 and 16 and legs 13. Find the height and area.*

   ANSWER $h = 12$, $S = 132$

5. *$MN \parallel BC$ with $AM = 4$, $MB = 8$ and $AN = 5$. Find NC.*

   ANSWER 10

6. *The angles of a quadrilateral are in the ratio $3 : 4 : 5 : 6$. Find them.*

   ANSWER $60°, 80°, 100°, 120°$

7. *A square has diagonal 12. Find its area.*

   ANSWER 72

Hard · stretch and reason

7 PROBLEMS

1. *Project 2, step 4: predict the interior and exterior sums for a 9-gon, then check.*

   ANSWER $1260°$ interior, $360°$ exterior.

2. *Project 2, step 5: explain in one sentence why the exterior sum does not depend on n.*

   ANSWER Walking once round the polygon turns you through exactly one full revolution, whatever the number of corners.

3. *Prove that the midpoints of the sides of a rhombus form a rectangle.*

   ANSWER Each side of the new quadrilateral is parallel to a diagonal, and the diagonals of a rhombus are perpendicular — so the new angles are right angles.

4. *A parallelogram has perimeter 40 and one diagonal 16, with the diagonals perpendicular. Find the sides and the other diagonal.*

   ANSWER Perpendicular diagonals make it a rhombus: side 10, half-diagonals 8 and 6, so the other diagonal is 12.

5. *In trapezium ABCD (AD ∥ BC), the midline is 10 and the height is 7. A diagonal splits it into two triangles — find their total area.*

   ANSWER $S = 10 \cdot 7 = 70$ — the two triangles together are the whole trapezium.

6. *A convex polygon has exactly three right angles and the rest equal. If n = 6, find the equal angles.*

   ANSWER $720° - 270° = 450°$ shared by 3 angles: $150°$ each.

7. *Explain why a non-convex polygon can break the exterior-angle argument.*

   ANSWER At a reflex vertex the turn is in the opposite direction, so the turns no longer add to a single full revolution.

07 Homework

Homework before the control work

Geometry 8, Тест 1, p. 33. Then revise every definition and property from Темы 1–11.

1. *Work Тест 1 on p. 33 in full.*

2. *Write out the three tests for a parallelogram, with a sketch each.*
3. *Write out the diagonal properties of the parallelogram, rectangle, rhombus and square in one table.*
4. *A regular polygon has interior angle 162°. Find n.*
5. Finish Project 2 and write your one-sentence explanation for step 5.

Control work 1 · Quadrilaterals

The Chapter I assessment: polygons, the four special quadrilaterals, Thales' theorem and midlines.

LESSON 18 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 13–14 | STAGE 9 · UNIT 5 CHECK

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the whole of Chapter I in 40 minutes.
- Produce a correctly reasoned proof, not just a numerical answer.
- Draw and label a figure for every question.

TEXTBOOK

National	pp. 33–34
Cambridge	Learner's Book pp. 103–126
Workbook	Workbook Unit 5

01 Explanation

The paper — 40 minutes, 7 tasks, 14 marks

Two variants, printed below as the Easy and Medium sets. The balance:

TASK	TOPIC	SOURCE	MARKS
1	Angle sum of a polygon	Тема 1	2
2	Angles of a parallelogram	Тема 2	2
3	Rectangle — diagonal or area	Тема 4	2
4	Rhombus or square	Темы 5–6	2
5	Trapezium — height or midline	Темы 7–8, 10–11	2
6	Thales' theorem	Тема 9	2
7	A proof	Темы 2–3	2

Marking note to give before the paper starts: a numerical answer with no figure and no reason earns one mark out of two. Task 7 earns nothing at all without a stated reason at each step — say which property or test you are using.

What to check while they write

- Is there a labelled figure? Learners who skip the figure lose marks on tasks 5 and 7 first.
- On task 6, has the proportion been written before any numbers were substituted?
- On task 7, is a **test** being quoted where a test is needed, rather than a property?

The Hard set below is the work-on-mistakes material: seven pieces of wrong reasoning drawn from the errors this paper produces every year.

02 Worked examples

EXAMPLE 1 **Find the error: “ABCD is a parallelogram with AC = BD, therefore it is a rhombus.”**

1. Equal diagonals is the **rectangle** condition, not the rhombus one.
2. The rhombus condition is **perpendicular** diagonals.
3. So $ABCD$ is a rectangle.
   It is a rhombus only if it is also a square.

ANSWER

A rectangle, not a rhombus.

EXAMPLE 2 **Find the error: “ABCD has AC = BD, therefore it is a rectangle.”**

1. The word “parallelogram” is missing from the hypothesis.
2. An isosceles trapezium also has equal diagonals.
   A counter-example.
3. Equal diagonals alone determine nothing.

ANSWER

Not necessarily a rectangle — it could be an isosceles trapezium.

03 Interactive model

Use this in the work-on-mistakes lesson: show the wrong reasoning, take a vote, then reveal.

Find the flaw in the reasoning

> **Claimed:** in parallelogram $ABCD$, $AC = BD$.

1. The parallelogram property is that the diagonals **bisect each other**. That is $AO = OC$ and $BO = OD$.

2. Nothing forces AC and BD to be the same length.

3. Drag any non-rectangular parallelogram and the two diagonals differ. Equality needs a right angle.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Figure (diagram)	Chizma	Чертёж
Reason (justification)	Asos	Обоснование
Error	Xato	Ошибка
Prove	Isbotlash	Доказать
Property	Xossa	Свойство
Test (criterion)	Alomat	Признак

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A parallelogram with perpendicular diagonals is:

- a rectangle
- a rhombus
- a trapezium
- nothing special

An isosceles trapezium has:

- equal diagonals
- perpendicular diagonals
- diagonals that bisect each other
- no special diagonals

The midline of a trapezium with bases 8 and 14 is:

- 7
- 11
- 22
- 3

To prove a quadrilateral is a parallelogram you need:

- a property
- a test
- either
- the angle sum

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Find the sum of the interior angles of a heptagon.*

   ANSWER $900°$

2. ***Task 2.*** *In parallelogram* $ABCD$, $\angle A = 72°$. *Find the other three angles.*

   ANSWER $108°, 72°, 108°$

3. ***Task 3.*** *A rectangle has sides 8 cm and 15 cm. Find the diagonal.*

   ANSWER 17 cm

4. ***Task 4.*** *A rhombus has diagonals 10 cm and 24 cm. Find its side and area.*

   ANSWER 13 cm, 120 cm^2

5. ***Task 5.*** *A trapezium has bases 7 cm and 13 cm. Find its midline.*

   ANSWER 10 cm

6. ***Task 6.*** $MN \parallel BC$, $AM = 3$, $MB = 6$, $AN = 4$. *Find* NC.

   ANSWER 8

7. ***Task 7.*** *Prove that a quadrilateral whose diagonals bisect each other is a parallelogram.*

   ANSWER $\triangle AOB \cong \triangle COD$ by SAS $\Longrightarrow AB = CD$ and $AB \parallel CD \Longrightarrow$ Test 1.

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *A regular polygon has exterior angle 30°. Find n and the interior sum.*

   ANSWER $n = 12$, $1800°$

2. ***Task 2.*** *In parallelogram ABCD,* $\angle A - \angle B = 30°$. *Find all four angles.*

   ANSWER $105°, 75°, 105°, 75°$

3. ***Task 3.*** *A rectangle has diagonal 25 cm and one side 7 cm. Find the area.*

   ANSWER $168\ cm^2$

4. ***Task 4.*** *A square has diagonal 10 cm. Find its side and area.*

   ANSWER $5\sqrt{2}$ cm, $50\ cm^2$

5. ***Task 5.*** *An isosceles trapezium has bases 5 cm and 17 cm and legs 10 cm. Find the height and area.*

   ANSWER $h = 8$ cm, $S = 88\ cm^2$

6. ***Task 6.*** $MN \parallel BC$, $AM = 4$, $AB = 10$, $AC = 15$. *Find AN.*

   ANSWER 6

7. ***Task 7.*** *Prove that a parallelogram with equal diagonals is a rectangle.*

   ANSWER Bisecting + equal $\Longrightarrow \triangle ABC \cong \triangle BAD$ by SSS $\Longrightarrow \angle A = \angle B = 90°$.

Hard · stretch and reason

7 PROBLEMS

1. *Find the flaw: "the diagonals of a parallelogram are equal".*

   ANSWER They bisect each other; equal needs a rectangle.

2. *Find the flaw: "$AC = BD$, therefore $ABCD$ is a rectangle".*

   ANSWER Only if $ABCD$ is already a parallelogram — an isosceles trapezium is a counter-example.

3. *Find the flaw: "$MN \parallel BC$, $AM = 3$, $AB = 12$, so $AN : NC = 3 : 12$".*

   ANSWER Part-to-whole mixed with part-to-part. Use $\frac{3}{12} = \frac{AN}{AC}$ or $\frac{3}{9} = \frac{AN}{NC}$.

4. *Find the flaw: "bases 8 and 14, so the midline is 7".*

   ANSWER That is the triangle rule. For a trapezium the midline is the average: 11.

5. *Find the flaw: "a rhombus is a rectangle because both are parallelograms".*

   ANSWER Both are parallelograms, but neither is a special case of the other — only the square is both.

6. *Find the flaw: "opposite sides equal, therefore a rectangle".*

   ANSWER Opposite sides equal gives a parallelogram (Test 2); a rectangle needs a right angle or equal diagonals as well.

7. *Find the flaw: "three lines evenly spaced by eye cut equal segments, by Thales".*

   ANSWER Thales' theorem requires the lines to be **parallel**, which "evenly spaced by eye" does not establish.

07 Homework

After the control work

Re-solve, in full, every task you lost marks on. Quarter II begins with Pythagoras' theorem.

1. *Re-solve every task you lost marks on, with a labelled figure and a stated reason.*
2. *Write out the four errors from the Hard set in your own words.*

3. *Revise the properties of a right-angled triangle from Grade 7 — the next chapter opens with Pythagoras.*